

MFE 230GA · FINAL PROJECT · FALL 2026

Follow the Workers: Results Book

Every result, figure, table and claim behind the report, with its source file

TEAM

Piero	PELOSI
Alex	ROESLER
Romain	ALMEIDA
Elouan	BAHRI
Al Yazid	BENSAID

COURSE

MFE 230GA – Equity Markets

INSTRUCTORS

Raffaele Savi and Gerald Garvey
BlackRock

GRADUATE STUDENT INSTRUCTOR

Vinicio DeSola

Contents

1	Verdict	2
2	Scoreboard	4
3	Figures	5
4	Statistics	12
5	HQ12 robustness battery (v3)	16
6	Study 1 robustness and Study 2.3	21
7	The agent experiment	25
8	AI-interaction inventory	26
9	Claims register	28
10	Known issues	33

1 Verdict

Verdict on each strategy

Strategy	Label	Verdict
A3, preregistered primary	CONFIRMATORY	<i>Do not implement.</i> Test Sharpe -0.58 , IC -0.021 ($t = -1.08$), placebo $p = 0.72$, fails gates G2–G5. Nothing below changes this. [C01–C07]
A0, A1, A1-T, A4	CONFIRMATORY	Negative in the test (-0.36 to -0.79); not primary; no claim. [C08]
Study 1b (Opus follow-up), A4	expl.	Launched after the test was seen. Sharpe $+0.01$ but IC $t = -0.10$, alpha $t = 0.05$, fails G2–G5; its gross came from one long industry. Evidence about agents, not the strategy. [X01, X02, X04]
W wage growth (declared headline)	POST-HOC	Modest and fragile: test Sharpe $+0.16$, IC $+0.049$ ($t = 2.04$), placebo $p = 0.02$, but alpha $t = 0.29$, a value/momentum tilt (HML -0.43 , Mom $+0.22$), Sharpe -1.71 in the last 18 months, DSR 0.02. Not distinguishable from noise after the looks taken. Forward-tested. [V_V2a_SR, V01, V11]
SC signed composite	POST-HOC	Negative (test -0.15); learned signs did not help. [V_V2b_SR, V06]
W+MOM wage + momentum	POST-HOC	Test $+0.35$, alpha $t = 1.36$, placebo $p = 0.26$, DSR 0.09; momentum supplies the return. Its ETF book has four legs. [V_V2c_SR, V05, F04]
HQ12 $-(\text{hires} + \text{quits})$, 12-month hold	POST-HOC	<i>The thesis result, fragile in time.</i> Positive in every window ($+0.50$ research, $+0.29$ test, $+0.35$ full, $+0.29$ last 18m); 12-month IC $+0.096$ ($t = 2.43$) against a 1-month IC of -0.02 ; alpha $+0.26\%$ a month ($t = 1.5$ test, 1.9 full); survives components, holding 9–18 months, 50 bp costs (break-even 68 bp) and every drop-one run (min $+0.18$). Fails one pre-declared condition: first test half -0.21 (second $+0.64$); the test bootstrap interval includes zero; DSR 0.05 at 132 trials. Not tradeable on this evidence; the forward test is live. [V_V2d_SR, V02, V03, R1a–R9a, R12, R13]

The story in one paragraph

Public labor flows do carry information about industry returns, but as slow cost news rather than fast demand news. The preregistered strategy traded the fast channel: it bet that industries with rising openings, hires and hours would outperform next month, and it lost money in a one-time test on 142 untouched months (Sharpe -0.58). Two of its four signs were wrong and the one stable short-horizon signal, wage growth, had been dropped during the build. When we look at horizons instead of months, the picture inverts: industries that hire and lose workers fastest earn lower returns over the following 9 to 24 months, as investment-based asset pricing predicts, and a 12-month book built on that sign (HQ12) is positive in every window we have. That book was designed after the test was opened, it was negative in the first half of the test, and its deflated Sharpe is near zero, so it is a hypothesis with a live forward test, not an implementable strategy. The verdict is still *Do not implement*.

What we can claim

- The preregistered strategy fails cleanly and we know why: wrong signs [D03], wrong horizon [D04, R2a], an untestable moderator [D06, D07], agent features active in 3 test months [D12], over-shrinkage and a binding risk cap [C18, D13], concentrated losses [D09]. Costs were not the cause [C15].
- Wage growth is the only short-horizon labor signal whose IC exceeds 1.5 standard errors in both periods [D01]; both blinded agent runs rediscovered it.
- Hiring and quits predict returns with a negative sign at 9–24 months and not at 1 month [R2a, R2b]; hires alone carry most of it [R1a, R1b].
- HQ12 is not carried by one industry [R7a], survives realistic costs [R5a] and longer holding

periods [R3a], and FF5 + momentum do not explain it (alpha t 1.5–1.9) [R6a, R6a2], but it loads against profitability [R6b].

- The 13-group universe caps what any IC can deliver: a 0.1 twelve-month IC with breadth 13 implies an IR near 0.35 at best; W realised far less than its IC implied [R11a].
- A stronger model followed the protocol better but found no more out-of-sample signal; the LLM judge over-attributed idea uptake [A01, A02, A03, D10].
- The forward test is frozen and the Q4 2026 books are dollar-neutral [F01–F05].

What we cannot claim

- That any post-hoc variant passed: none met the v2 reading rule [V07]; HQ12 fails R4 and its test bootstrap interval includes zero [R9a, R13]; every DSR is below 0.2 [V08, R12].
- That HQ12 is tradeable, or that the tightness pattern in R8 is more than one draw with 54–88 months per cell [R8a].
- That Studies 2.1 and 2.2 as first run reflect real-time data: those numbers use revised data with fixed lags [S02]; the as-known rerun came later, in Study 2.3 [T02].
- That the Opus study's +0.01 is evidence for the strategy [X01].
- Alpha from the labor signal beyond quality/growth exposure for A3 or W [C13, V11].

Figure plan for the report

1 fig_timeline [D06], 2 fig_agents [A01, A02], 3 fig_loadings [C13, V11, R6b], 4 fig_cumulative [C08, W], 5 fig_ic_signals [D01–D03], 6 fig_horizon [D04, D05], 7 fig_agents_scatter [D10–D12], 8 fig_horizon_profile [R2a, R2b], 9 fig_v2_windows [W–HQ12 Sharpe], 10 fig_v2d_dropone [R7a]; appendix: fig_tightness, fig_v2d_tightness, fig_industry_pnl, fig_v2d_holding.

Study 2.3: what changed in the reading

Run on 4 October 2026 from a spec hashed before running (analysis/specs/study2_3_robustness.md, 162 looks). Nothing changes in the verdict: Study 1's primary A3 stays "Do not implement" and HQ12 stays a post-hoc hypothesis that fails its reading rule. What the new runs add:

- *As-known data (R1)*. The four variants were rerun on data as known at each decision date, after the as-known inputs were validated to machine precision against Study 1's saved features and A0 scores. HQ12 keeps its sign and most of its size: test Sharpe +0.20 against +0.29 with fixed lags, 12-month IC +0.085 (t 2.26) against +0.096; its first test half is still negative (-0.23), its break-even cost falls to 50 bp, its drop-one minimum to +0.05 and its deflated Sharpe is 0.02 at 162 looks. W earns +0.13 and W+MOM +0.46. SC flips from -0.15 to +0.24 because three of its six learned signs change with the data version, so SC is not a stable object. As-known and fixed-lag HQ12 scores pick the same top three groups in only 39% of test months. The reading fixed in the spec is "as-known confirms the fixed-lag reading"; the months affected by the 2025 shutdown are handled correctly in this rerun. [T01–T09]
- *Risk overlay (R2)*. Reaching the 10% volatility target would need a median gross exposure of 2.6 for A3 and 2.4 for HQ12, so the cap of 2 binds in 87% and 83% of test months. With the cap at 3 or removed, realised volatility rises to about 10% and the Sharpe ratio does not improve (A3 -0.61 and -0.62; HQ12 +0.24). The cap explains the volatility shortfall, not the losses. [B05, T10, T11]
- *Rank-weighted books (R3)*. Holding all 13 groups in proportion to their rank changes little: A3 -0.49 (transfer coefficient 0.83) against -0.58; HQ12 +0.31 against +0.29. Portfolio construction is not what failed. [B06, T12]

- *Study 1 diagnostics*. The block-bootstrap 90% interval of A3's test Sharpe is [-0.99, -0.16], and the A3 score has no IC at any horizon from 1 to 24 months. Study 1's freeze record checks out: the hash of FREEZE.json equals the one stored when the test was opened, 0.04 seconds later. [B01-B04]
- *Agents (R4, R5)*. In both agent studies the gap between the communicating and independent arms is within the spread across seeds (permutation p 0.30 and 0.10, the smallest value 20 relabelings allow), and tokens per call are similar across arms. [T13, T15]
- *Breadth and capacity (R6, R7)*. The 13 groups behave like about 9 independent bets (participation ratio of relative returns 9.1). With the proxy ETFs of the forward test, monthly trading reaches 1% of one day's volume in the least liquid ETF at about \$0.5 million of AUM for A3 and \$1.2 million for HQ12; gate G5 stays failed. [T16, T17]
- *Judge fix (R8)*. Requiring a verbatim quote made the Sonnet judge's answers usable (8 of 8, 7 in agreement with the human) but did not reduce the Opus judge's false yes (8 of 17 against 7). [T18, T19]
- *Feedback ablation (R9)*. On research data only, 4 of 48 valid proposals used a regime gate when the feedback showed IC by tightness tercile and 3 of 50 when it did not. The rerun does not support the explanation that the tercile feedback caused the gates, and it produced fewer gates than the original run (15% to 29% of proposals per seed). [T14, T20, T21]

2 Scoreboard

The A3 sealed ledger starts from an empty book in November 2014; the v2 test figures are the test slice of a continuous run that starts in May 2004 (pre-declared in v2_spec.md section 4). IC is one-month except HQ12 (12-month, its holding horizon). DSR is the research-period value with 41 nominal trials for the frozen studies and the test-window value with 112 trials for v2. Labels: conf. = confirmatory, expl. = exploratory follow-up. Source: [results/scoreboard.csv](#).

Strategy	Label	Res.	Test	Full	18m	IC (t)	α/m (t)	Plac.	DSR	Verdict
Study 1: A0 fixed rule	conf.	-0.44	-0.36	-0.48	-1.56	-0.020 (-0.71)	-0.25% (-0.92)		0.00	negative, not primary
Study 1: A1 ridge	conf.	-0.54	-0.51	-0.52	+0.73	-0.014 (-0.74)	-0.35% (-1.72)		0.00	negative, not primary
Study 1: A1-T ridge x tightness	conf.	-0.55	-0.59	-0.58	-0.37	-0.029 (-1.42)	-0.42% (-2.06)		0.00	negative, not primary
Study 1: A3 agent features, independent (primary)	conf.	-0.13	-0.58	-0.44	+0.85	-0.021 (-1.08)	-0.38% (-1.86)	0.72	0.00	Do not implement
Study 1: A4 agent features, communicating	conf.	-0.18	-0.79	-0.56	+1.23	-0.020 (-0.86)	-0.55% (-2.39)		0.00	negative, not primary
Study 1b: A3 agent features, independent	expl.	-0.48	-0.46	-0.46	-0.64	+0.001 (+0.05)	-0.24% (-1.54)		0.00	exploratory, negative
Study 1b: A4 agent features, communicating (its primary)	expl.	-0.40	+0.01	-0.13	-1.41	-0.002 (-0.10)	+0.01% (+0.05)	0.56	0.00	exploratory, fails G2-G5
Study 2.1: W wage growth (declared headline)	post-hoc	+0.09	+0.16	+0.15	-1.71	+0.049 (+2.04)	+0.06% (+0.29)	0.02	0.02	noise after the looks taken
Study 2.1: SC signed composite	post-hoc	-0.06	-0.15	-0.11	-1.34	+0.042 (+1.65)	-0.18% (-0.84)	0.05	0.00	noise after the looks taken
Study 2.1: W+MOM wage growth + momentum	post-hoc	+0.27	+0.35	+0.32	-0.78	+0.032 (+1.21)	+0.22% (+1.36)	0.26	0.09	noise after the looks taken
Study 2.1: HQ12 hires + quits reversal, 12-month hold	post-hoc	+0.50	+0.29	+0.35	+0.29	+0.096 (+2.43)	+0.26% (+1.37)	0.85	0.06	noise after the looks taken
Study 2.3: W, as-known data	post-hoc	+0.11	+0.13	+0.13	-1.31	+0.041 (+1.73)	(+0.18)		0.01	noise after the looks taken
Study 2.3: SC, as-known data	post-hoc	+0.33	+0.24	+0.28	+1.38	+0.021 (+0.75)	(+0.71)		0.03	noise after the looks taken
Study 2.3: W+MOM, as-known data	post-hoc	+0.25	+0.46	+0.37	-0.94	+0.036 (+1.40)	(+1.93)		0.13	noise after the looks taken
Study 2.3: HQ12, as-known data	post-hoc	+0.54	+0.20	+0.31	+0.14	+0.085 (+2.26)	(+0.92)		0.02	noise after the looks taken

3 Figures

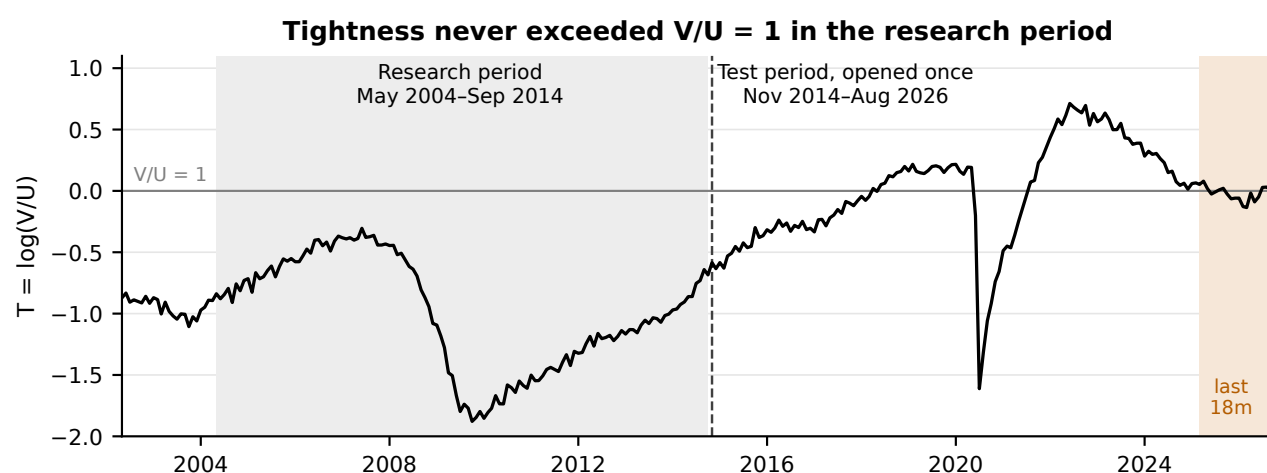


FIGURE 1. Tightness never exceeded $V/U = 1$ in the research period, so the tightness moderator could not be learned there. Source: analysis/data/fred (JTSJOL, UNEMPLOY). POST-HOC

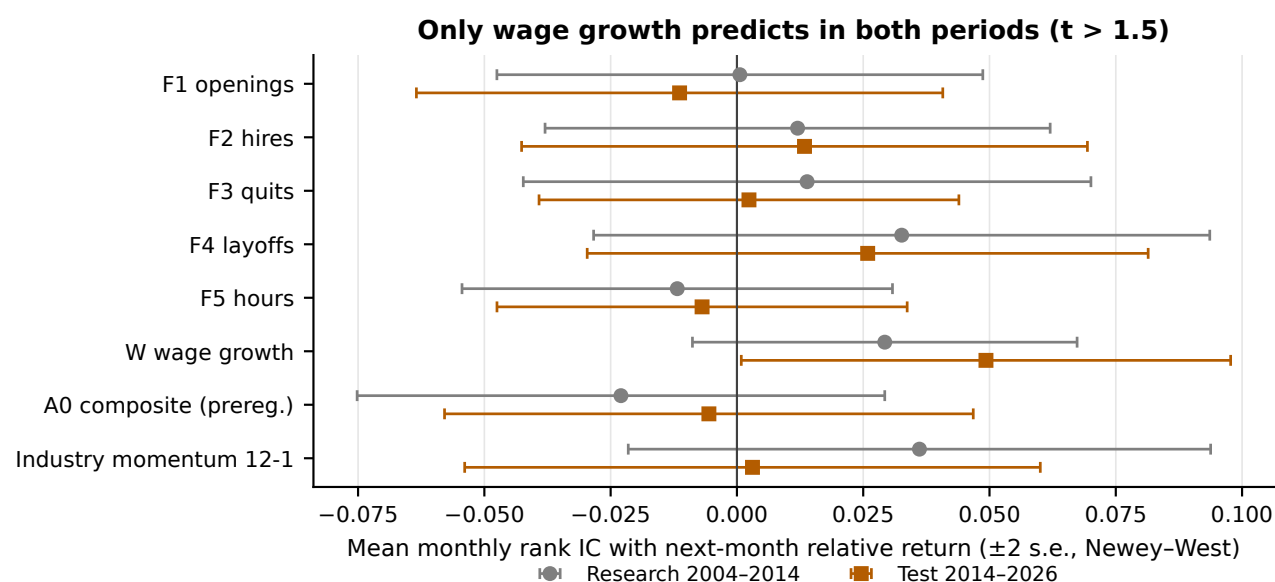


FIGURE 2. Only wage growth has an IC above 1.5 standard errors in both periods; layoffs and hours carry the opposite sign to the one A0 imposed. Source: analysis/output/study1/tables/d1_signal_ic.csv. POST-HOC

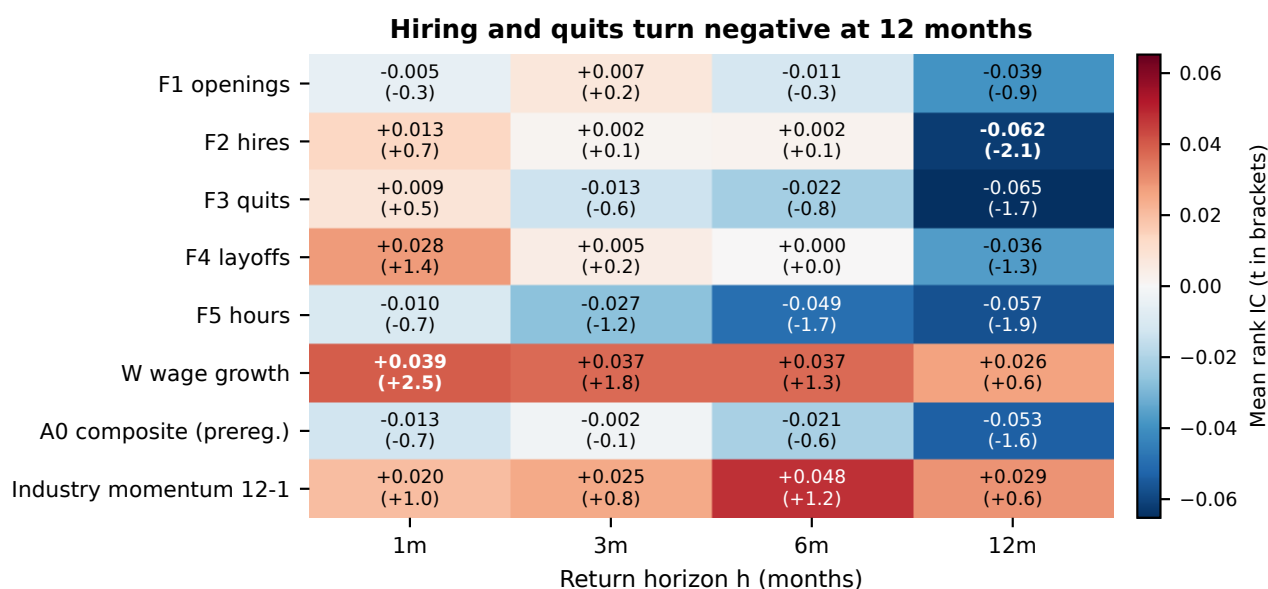


FIGURE 3. Hires and quits turn negative at 12 months (t -2.1, -1.7): the investment/cost channel shows up slowly, not at one month. Source: analysis/output/study1/tables/d2_horizon_ic.csv. POST-HOC

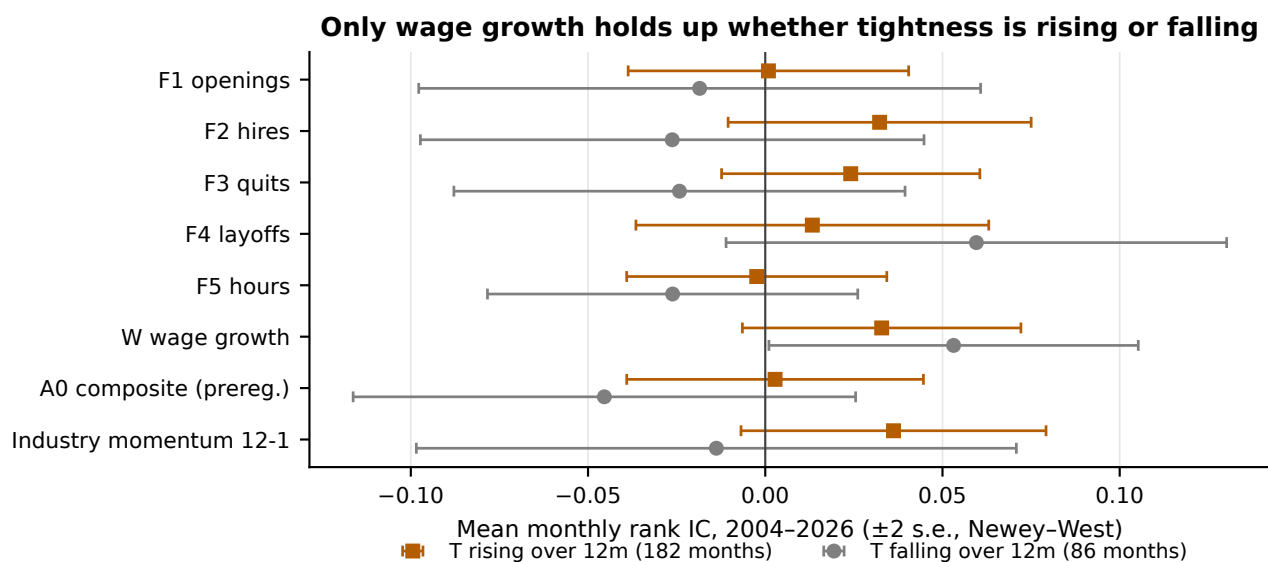


FIGURE 4. Split by whether T rose or fell over 12 months, hires and quits predict returns only while the market tightens; the split is balanced in both periods. Source: analysis/output/study1/tables/d3_tightness_ic.csv. POST-HOC

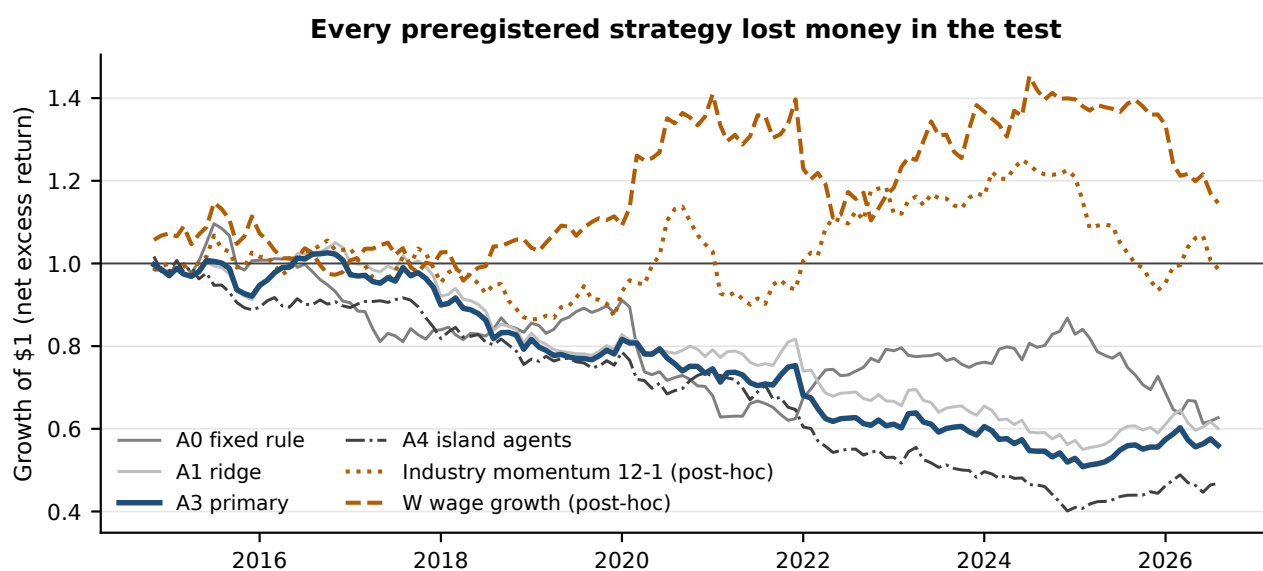


FIGURE 5. Every preregistered strategy lost money in the test; the post-hoc wage-growth book gained until 2024 and gave much of it back. Source: sealed/initial/ASOF_*)_ledger.csv; analysis/output/study2/tables/v2_monthly_net.csv.

CONFIRMATORY (A0-A4) / POST-HOC (momentum, W)

Losses: health care, construction

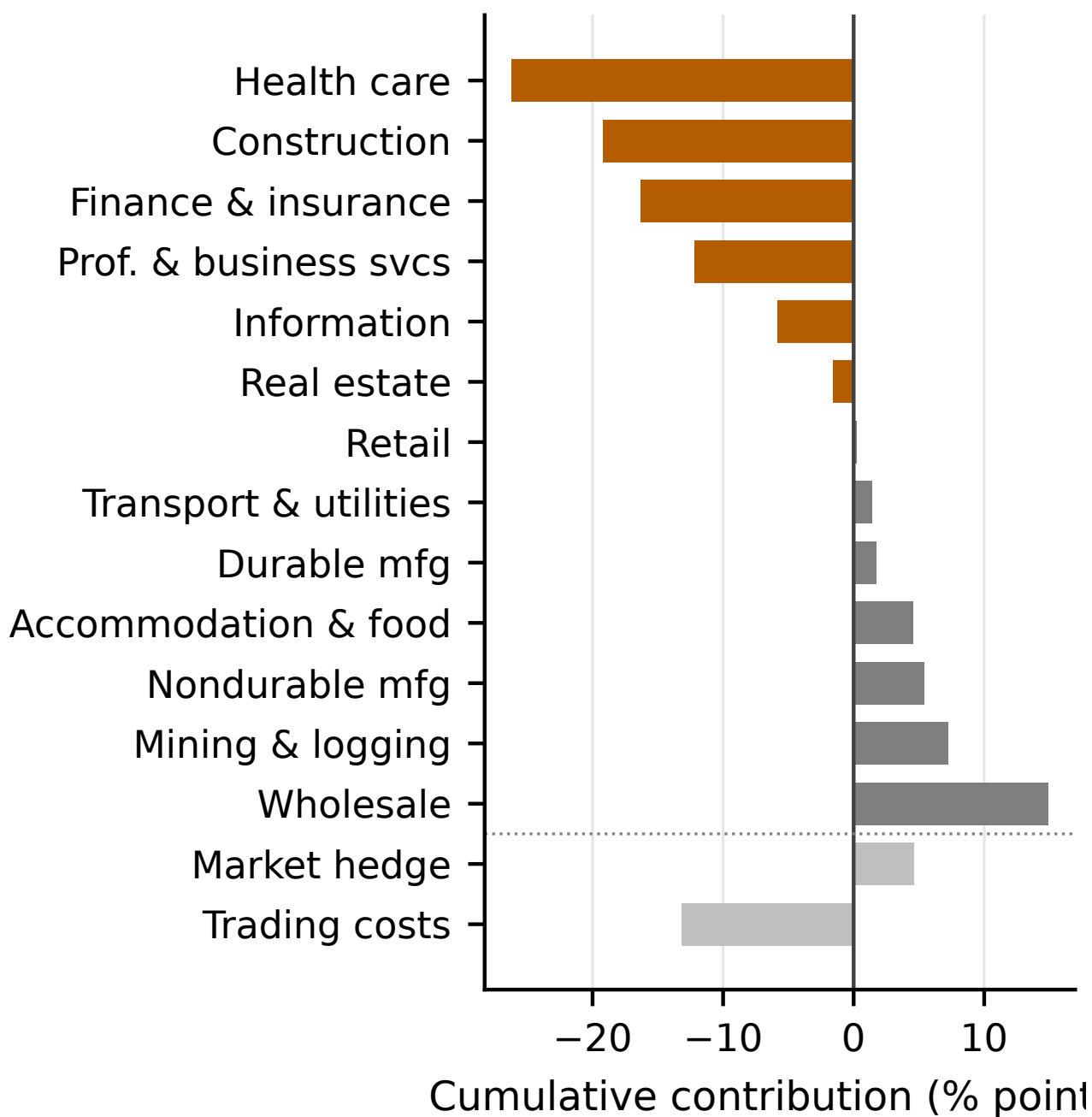


FIGURE 6. A3 lost most in health care, construction and finance; trading costs were a secondary drag. Source: analysis/output/study1/tables/d5_industry_pnl.csv.

POST-HOC

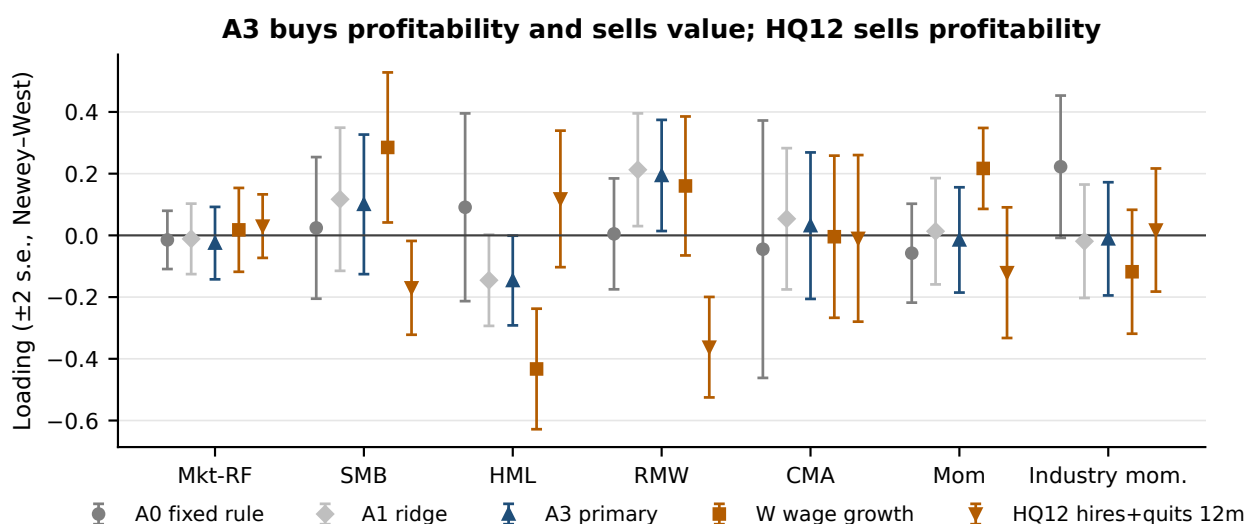


FIGURE 7. The labor books are quality/growth tilts: A3 loads on RMW and against HML; W loads against HML and on momentum. Source: results.json attribution; analysis/output/study2/tables/v2a_factor_loadings_test.csv.

CONFIRMATORY (A0, A1, A3) / POST-HOC (W)

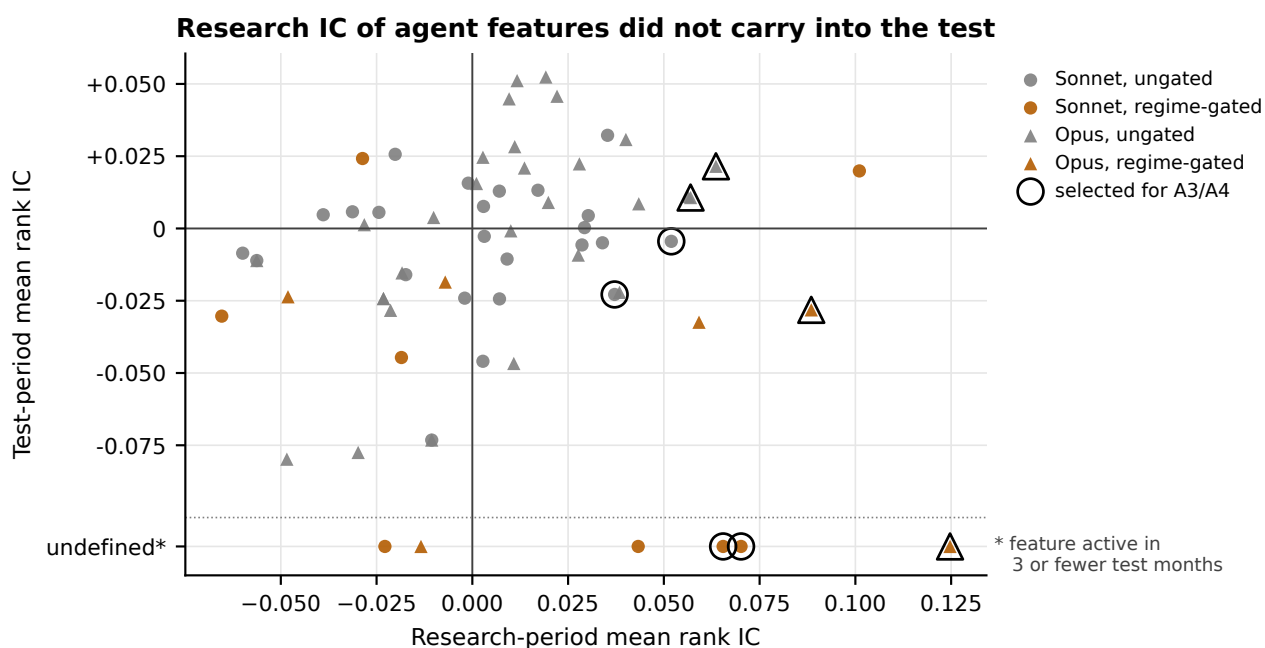


FIGURE 8. Research IC of agent features barely predicts their test IC; the selected regime-gated features were active in 3 test months. Source: analysis/output/study1/tables/d7_agent_candidates.csv.

POST-HOC

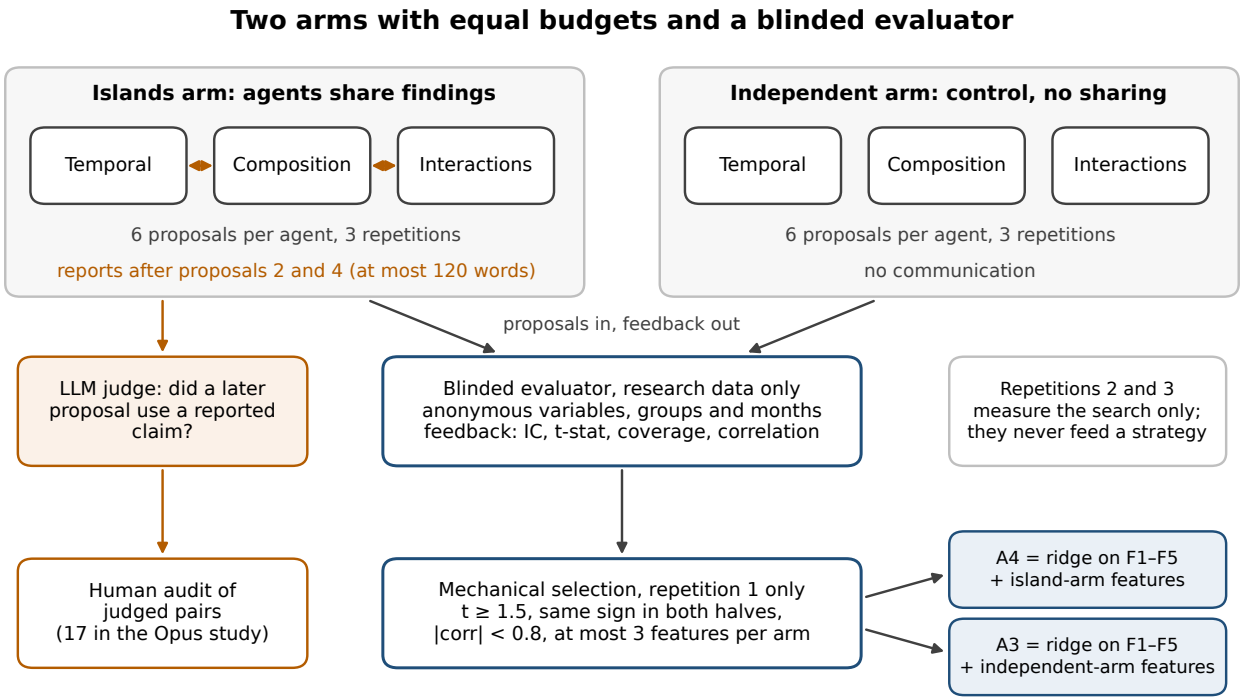


FIGURE 9. Two arms with equal budgets, a blinded evaluator, mechanical selection and a human-audited LLM judge. Source: frozen params.py AGENT. CONFIRMATORY

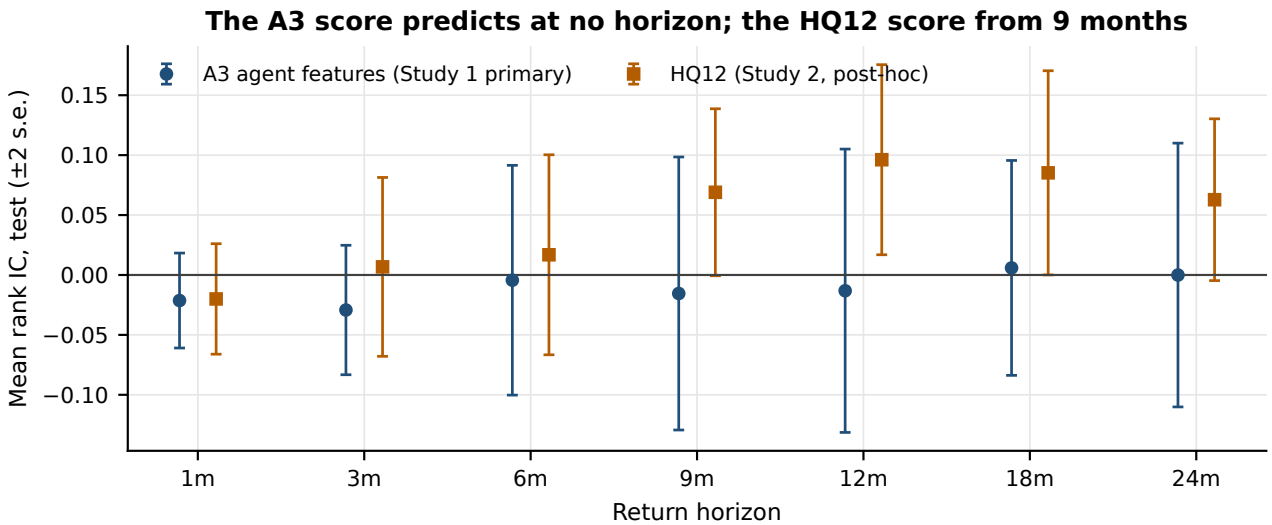


FIGURE 10. The A3 score has no IC at any horizon from 1 to 24 months; the HQ12 score has one from 9 months. Source: analysis/output/study1/tables/d11_a3_horizon_profile.csv. POST-HOC

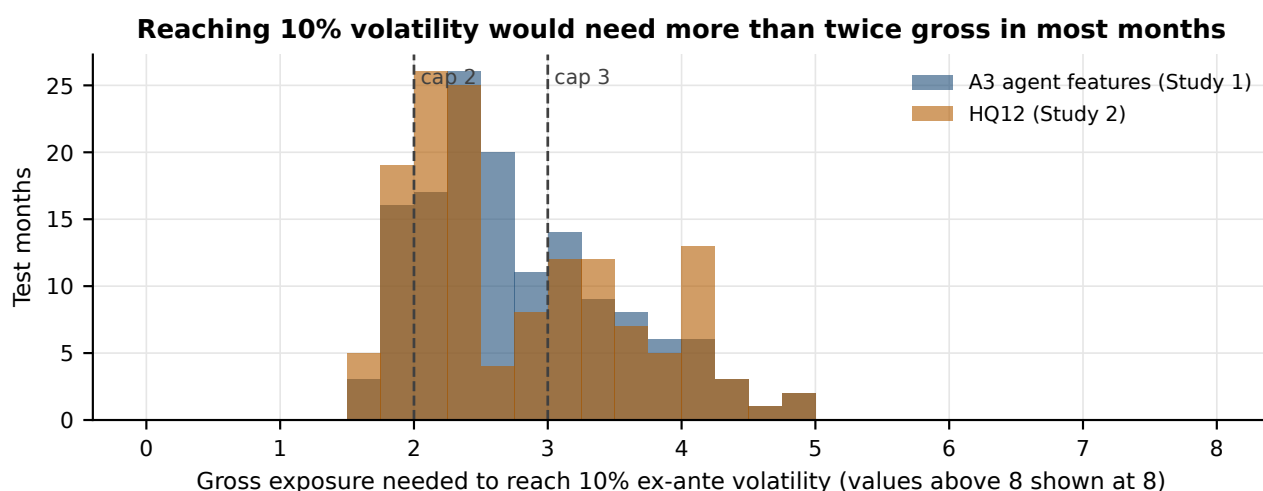


FIGURE 11. Reaching the 10% volatility target would need a gross exposure above 2 in most test months, so the cap binds; the cap explains the volatility shortfall, not the losses. Source: analysis/output/study2/tables/s23_r2_distribution.csv.

POST-HOC

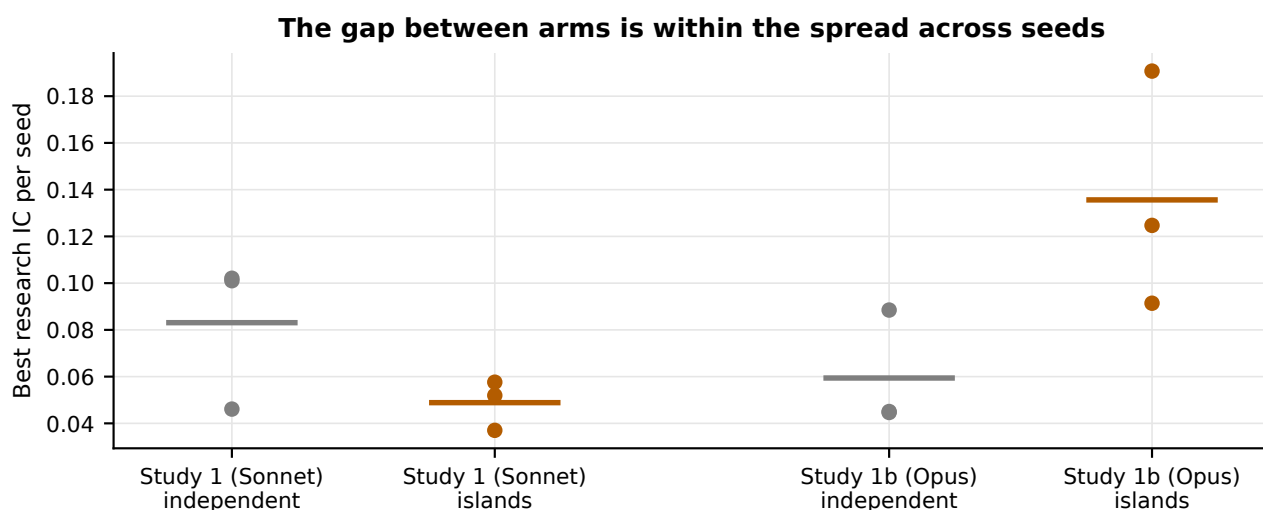


FIGURE 12. Best research IC per seed, by study and arm: the gap between arms is within the spread across seeds. Source: analysis/output/study2/tables/s23_r4_agents_by_seed.csv.

POST-HOC

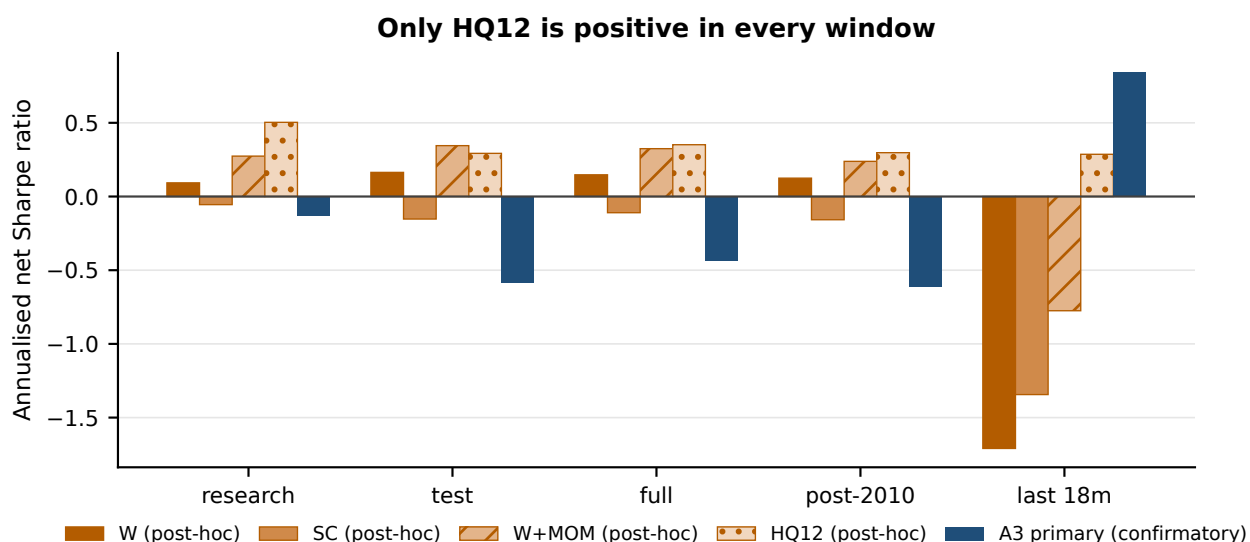


FIGURE 13. HQ12 is the only variant positive in every window; W and W+MOM lost heavily in the last 18 months. Source: analysis/output/study2/tables/v2_window_stats.csv.

POST-HOC / CONFIRMATORY (A3 bars)

4 Statistics

Each table lives in `analysis/output/study2/tables/<name>.csv`; the LaTeX fragment in `report/tables/<name>.tex`.

D1 single-signal rank IC

post-hoc

Signal	Research 2004–14	Test 2014–26	Full 2004–26
F1 openings	+0.001 (+0.02)	−0.011 (−0.44)	−0.005 (−0.30)
F2 hires	+0.012 (+0.48)	+0.013 (+0.48)	+0.013 (+0.71)
F3 quits	+0.014 (+0.49)	+0.002 (+0.11)	+0.009 (+0.50)
F4 layoffs	+0.033 (+1.07)	+0.026 (+0.93)	+0.028 (+1.38)
F5 hours	−0.012 (−0.55)	−0.007 (−0.34)	−0.010 (−0.67)
W wage growth	+0.029 (+1.54)	+0.049 (+2.04)	+0.039 (+2.50)
A0 composite (prereg.)	−0.023 (−0.88)	−0.006 (−0.21)	−0.013 (−0.69)
Industry momentum 12-1	+0.036 (+1.25)	+0.003 (+0.11)	+0.020 (+0.99)

D2 IC by horizon

post-hoc

Signal	1m	3m	6m	12m
F1 openings	−0.005 (−0.31)	+0.007 (+0.23)	−0.011 (−0.28)	−0.039 (−0.92)
F2 hires	+0.013 (+0.70)	+0.002 (+0.07)	+0.002 (+0.08)	−0.062 (−2.08)
F3 quits	+0.009 (+0.50)	−0.013 (−0.55)	−0.022 (−0.78)	−0.065 (−1.73)
F4 layoffs	+0.028 (+1.37)	+0.005 (+0.17)	+0.000 (+0.01)	−0.036 (−1.28)
F5 hours	−0.010 (−0.67)	−0.027 (−1.24)	−0.049 (−1.69)	−0.057 (−1.86)
W wage growth	+0.039 (+2.51)	+0.037 (+1.77)	+0.037 (+1.28)	+0.026 (+0.63)
A0 composite (prereg.)	−0.013 (−0.69)	−0.002 (−0.09)	−0.021 (−0.61)	−0.053 (−1.58)
Industry momentum 12-1	+0.020 (+0.99)	+0.025 (+0.80)	+0.048 (+1.24)	+0.029 (+0.63)

D3a tightness regime balance

post-hoc

Months in regime	Research 2004–14	Test 2014–26
months	125	142
expanding low	49	3
expanding middle	26	6
expanding high	50	133
T rising (12m)	93	88
T falling (12m)	32	54
V/U > 1	0	75
T min	−1.88	−1.61
T max	−0.30	+0.71

D3b IC when T rises vs falls

post-hoc

Signal	T rising (182 m)	T falling (86 m)	Difference
F1 openings	+0.001 (+0.05)	−0.019 (−0.47)	+0.019
F2 hires	+0.032 (+1.51)	−0.026 (−0.74)	+0.059
F3 quits	+0.024 (+1.32)	−0.024 (−0.76)	+0.048
F4 layoffs	+0.013 (+0.53)	+0.060 (+1.69)	−0.046
F5 hours	−0.002 (−0.13)	−0.026 (−1.00)	+0.024
W wage growth	+0.033 (+1.67)	+0.053 (+2.04)	−0.020
A0 composite (prereg.)	+0.003 (+0.13)	−0.045 (−1.28)	+0.048
Industry momentum 12-1	+0.036 (+1.68)	−0.014 (−0.33)	+0.050

D4 test-period performance

confirmatory (A0-A4) / post-hoc (momentum)

strategy	Sharpe	Net p.a.	Vol.	Max DD	Turnover	Costs p.a.
A0 fixed rule	−0.36	−3.5%	9.7%	−40.3%	0.83	1.01%
A1 ridge	−0.51	−4.0%	7.9%	−37.7%	0.91	1.11%
A3 primary	−0.58	−4.6%	7.8%	−41.1%	0.91	1.11%
A4 island agents	−0.79	−6.1%	7.8%	−53.3%	1.11	1.35%
Industry momentum 12-1	+0.03	+0.3%	9.4%	−20.8%	0.53	0.65%

D5 A3 P&L by industry

post-hoc

component	Contribution (pp)	Months long	Months short
Mining & logging	+7.2	33%	36%
Construction	−19.2	36%	39%
Durable mfg	+1.8	31%	24%
Nondurable mfg	+5.4	32%	35%
Wholesale	+14.9	31%	42%
Retail	+0.2	39%	35%
Transport & utilities	+1.4	47%	30%
Information	−5.8	30%	37%
Finance & insurance	−16.3	34%	45%
Real estate	−1.6	24%	45%
Prof. & business svcs	−12.2	18%	47%
Health care	−26.2	41%	36%
Accommodation & food	+4.5	46%	22%
Market hedge	+4.6		
Trading costs	−13.1		

D6 factor loadings

confirmatory

Factor	A0 fixed rule	A1 ridge	A3 primary
Alpha (per month)	−0.25% (−0.92)	−0.35% (−1.72)	−0.38% (−1.86)
Market	−0.01 (−0.31)	−0.01 (−0.20)	−0.03 (−0.43)
SMB	+0.02 (+0.21)	+0.12 (+1.01)	+0.10 (+0.89)
HML	+0.09 (+0.60)	−0.15 (−1.97)	−0.15 (−2.01)
RMW	+0.00 (+0.05)	+0.21 (+2.33)	+0.19 (+2.16)
CMA	−0.04 (−0.21)	+0.05 (+0.47)	+0.03 (+0.26)
Momentum	−0.06 (−0.72)	+0.01 (+0.15)	−0.01 (−0.17)
Industry momentum	+0.22 (+1.93)	−0.02 (−0.21)	−0.01 (−0.12)

D7a agent candidates

post-hoc

Repetition-1 candidates	Sonnet	Opus
valid candidates	33.0	35.0
regime-gated	9.0	6.0
test IC undefined (≤ 3 active months)	4.0	2.0
corr(research IC, test IC)	0.22	0.37
passed research filter	4.0	7.0
selected	4.0	4.0
selected and regime-gated	2.0	2.0
selected with positive test IC	0.0	2.0

D7b selected agent features

post-hoc

Study	Arm	Feature (decoded)	Research IC (n)	Test IC (n)
Sonnet	independent	regime(yoy(Earnings), "low")	+0.065 (48)	n/a (3)
Sonnet	independent	regime(spread(tsz(Layoffs, 60), tsz(Hires, 60)), "low")	+0.070 (43)	n/a (3)
Sonnet	islands	spread(chg(Layoffs, 3), chg(Openings, 3))	+0.037 (123)	−0.023 (142)
Sonnet	islands	spread(chg(Layoffs, 6), chg(Openings, 6))	+0.052 (120)	−0.004 (142)
Opus	independent	mult(regime(tsz(Hires, "exp"), "high"), cstrank(tsz(Openings, "exp")))	+0.088 (15)	−0.028 (133)
Opus	islands	tsz(lchg(lag(Earnings, 1), 12), 36)	+0.064 (78)	+0.022 (142)
Opus	islands	regime(spread(chg(Quits, 3), chg(Hires, 3)), "low")	+0.125 (48)	n/a (3)
Opus	islands	tsz(lchg(lag(Hours, 1), 12), 36)	+0.057 (78)	+0.011 (142)

D8 A3 by window

confirmatory

window	Months	Sharpe	Net return p.a.	Mean IC (t)
Full test, Nov 2014-Aug 2026	142	−0.58	−4.6%	−0.021 (−1.08)
First half	71	−0.72	−4.9%	−0.028 (−0.90)
Second half	71	−0.49	−4.3%	−0.015 (−0.59)
Excluding Mar-Dec 2020	132	−0.51	−4.1%	−0.012 (−0.56)
Last 18 months (Mar 2025-Aug 2026)	18	+0.85	+6.7%	+0.033 (+0.44)
Course window "full" (from Jun 2007)	231	−0.44	−3.2%	−0.009 (−0.49)
Course window "post-2010"	200	−0.62	−4.4%	−0.019 (−1.02)

D9 risk overlay

post-hoc

Study, strategy	Cap binds	Ex-ante vol	Realised vol	Gross leverage	Hedge
Sonnet A0	80%	8.0%	9.7%	1.96	0.15
Sonnet A1	88%	7.5%	7.9%	1.98	0.13
Sonnet A1-T	82%	7.5%	8.2%	1.97	0.12
Sonnet A3	87%	7.5%	7.8%	1.98	0.13
Sonnet A4	91%	7.3%	7.8%	1.99	0.13
Opus A3	92%	7.3%	7.5%	1.99	0.15
Opus A4	93%	7.3%	8.4%	1.99	0.11

D10 agent experiment design

confirmatory

study	item	value
Both	Arms	islands (communicating), independent (control)
Both	Agents per arm	Temporal, Composition, Interactions
Both	Proposals per agent x repetitions	6 x 3
Both	Migration reports	after proposals 2 and 4, at most 120 words
Both	Selection (repetition 1 only)	$t \geq 1.5$, same sign in both halves, $ \text{corr} < 0.8$, at most 3 per arm
Sonnet	Valid proposals	98 / 108
Sonnet	Migration reports admitted	7 / 18
Sonnet	Defined LLM-judge uptake labels	8 / 36
Sonnet	Human-audited judge pairs	8
Opus	Valid proposals	106 / 108
Opus	Migration reports admitted	15 / 18
Opus	Defined LLM-judge uptake labels	84 / 88
Opus	Human-audited judge pairs	17
Opus	Human-judge agreement	10 / 17 (human yes 3, judge yes 10)

v2 variants: Sharpe by window and test statistics**post-hoc**

Variant	Research	Test	Full	Post-2010	Last 18m	Test IC (t)	Alpha t	Placebo p	DSR
W	+0.09	+0.16	+0.15	+0.12	−1.71	+0.049 (+2.04)	+0.29	0.02	0.02
SC	−0.06	−0.15	−0.11	−0.16	−1.34	+0.042 (+1.65)	−0.84	0.05	0.00
W+MOM	+0.27	+0.35	+0.32	+0.24	−0.78	+0.032 (+1.21)	+1.36	0.26	0.09
HQ12	+0.50	+0.29	+0.35	+0.30	+0.29	−0.020 (−0.87)	+1.37	0.85	0.06

5 HQ12 robustness battery (v3)

Spec analysis/specs/study2_2_hq12_robustness.md frozen 2026-10-03T09:52:27Z, first run 09:53:32Z; every check run once; N_{trials} raised to 132. Results: analysis/output/study2/v3_results.md.

POST-HOC

Reading against the pre-declared rule: fragile in one dimension (R_4). HQ12 lost money in the first half of the test (Sharpe -0.21) and made it in the second ($+0.64$); the other six conditions hold. The 12-month IC is positive in both halves.

R1 Components

post-hoc

Sharpe research	Sharpe test	Sharpe full	Test IC 12m (t)	Test alpha t
+0.41	+0.19	+0.27	+0.093 (+2.90)	+0.79
+0.18	+0.16	+0.15	+0.085 (+1.75)	+0.81
+0.10	+0.42	+0.29	+0.082 (+1.38)	+1.48
+0.11	+0.16	+0.13	+0.032 (+0.87)	+0.69
+0.50	+0.29	+0.35	+0.096 (+2.43)	+1.37

R2 Horizon profile

post-hoc

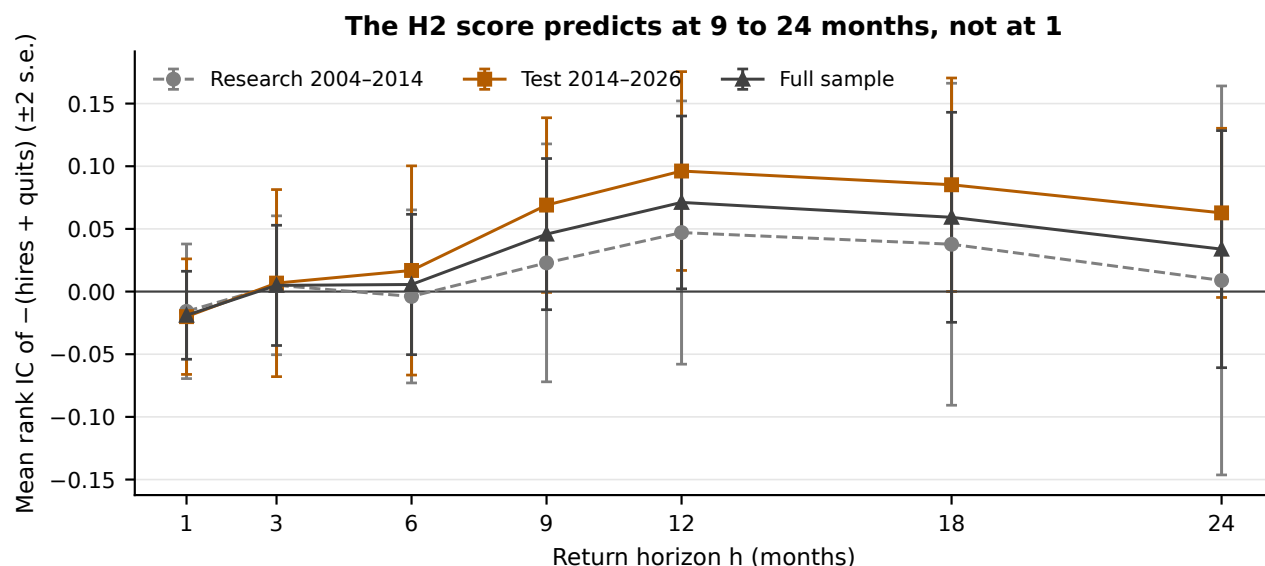


FIGURE 14. The HQ12 score has no one-month IC and a positive IC from 9 to 24 months, peaking at 12 (test $+0.096$, $t = 2.43$). Source: analysis/output/study2/tables/v3_r2_horizon_profile.csv.

POST-HOC

research	test	full
-0.016 (-0.59)	-0.020 (-0.87)	-0.019 (-1.08)
+0.005 (+0.18)	+0.007 (+0.18)	+0.005 (+0.21)
-0.004 (-0.11)	+0.017 (+0.40)	+0.006 (+0.20)
+0.023 (+0.48)	+0.069 (+1.98)	+0.046 (+1.52)
+0.047 (+0.90)	+0.096 (+2.43)	+0.071 (+2.06)
+0.038 (+0.59)	+0.085 (+2.00)	+0.059 (+1.41)
+0.009 (+0.11)	+0.063 (+1.86)	+0.034 (+0.72)

R3 Holding period

post-hoc

Positive at every holding period, 6 to 18 m

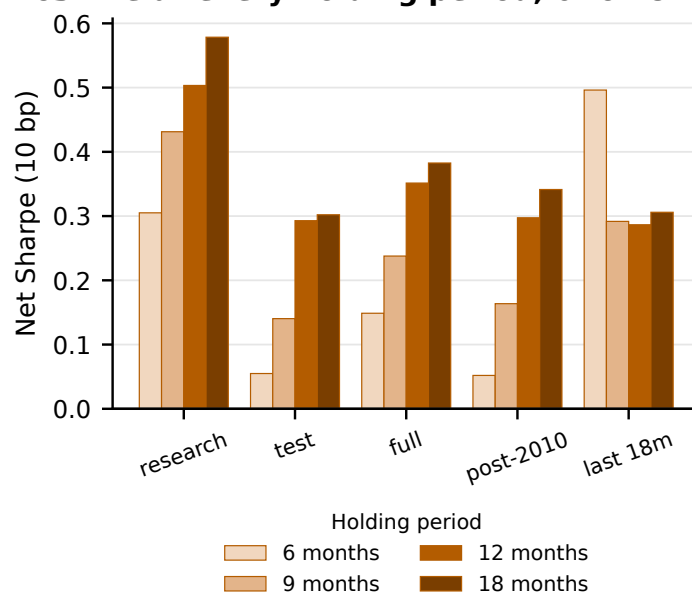


FIGURE 15. Longer holding helps: test Sharpe +0.05 at 6 months, +0.29 at 12, +0.30 at 18. Source: v3_r3_holding.csv.

POST-HOC

Holding	research	test	full	post-2010	last 18m	Test turnover
6 months	+0.31	+0.05	+0.15	+0.05	+0.50	0.49
9 months	+0.43	+0.14	+0.24	+0.16	+0.29	0.40
12 months	+0.50	+0.29	+0.35	+0.30	+0.29	0.37
18 months	+0.58	+0.30	+0.38	+0.34	+0.31	0.28

R4 Sub-periods

post-hoc

Window	Months	Sharpe	Net p.a.	IC 12m (t)
test: first half	71	-0.21	-1.4%	+0.097 (+1.75)
test: second half	71	+0.64	+6.5%	+0.095 (+1.60)
test: excl. Mar 2020-Dec 2021	120	+0.37	+3.0%	+0.105 (+2.23)
test: pre-2020	62	+0.14	+0.8%	+0.114 (+1.92)
test: 2020 onward	80	+0.37	+3.9%	+0.080 (+1.54)
full: first half	134	+0.33	+2.0%	+0.038 (+0.73)
full: second half	134	+0.38	+3.3%	+0.107 (+2.66)
full: excl. Mar 2020-Dec 2021	246	+0.40	+2.9%	+0.073 (+1.93)
full: pre-2020	188	+0.36	+2.2%	+0.068 (+1.57)

R5 Costs

post-hoc

Window	0 bp	10 bp	25 bp	50 bp	Break-even (bp)
research	+0.58	+0.50	+0.39	+0.20	76
test	+0.34	+0.29	+0.22	+0.09	68
full	+0.41	+0.35	+0.26	+0.11	69
post-2010	+0.35	+0.30	+0.21	+0.07	63
last 18m	+0.32	+0.29	+0.24	+0.16	99

R6 Factor regression, 12 HAC lags

post-hoc

Test (HAC 12)	Full (HAC 12)
+0.26% (+1.48)	+0.26% (+1.92)
+0.03 (+0.58)	−0.00 (−0.08)
−0.17 (−2.24)	−0.02 (−0.31)
+0.12 (+1.07)	+0.10 (+1.23)
−0.36 (−4.45)	−0.16 (−1.83)
−0.01 (−0.07)	−0.09 (−0.87)
−0.12 (−1.14)	−0.03 (−0.51)
+0.02 (+0.17)	+0.06 (+0.93)

R7 Drop one industry

post-hoc

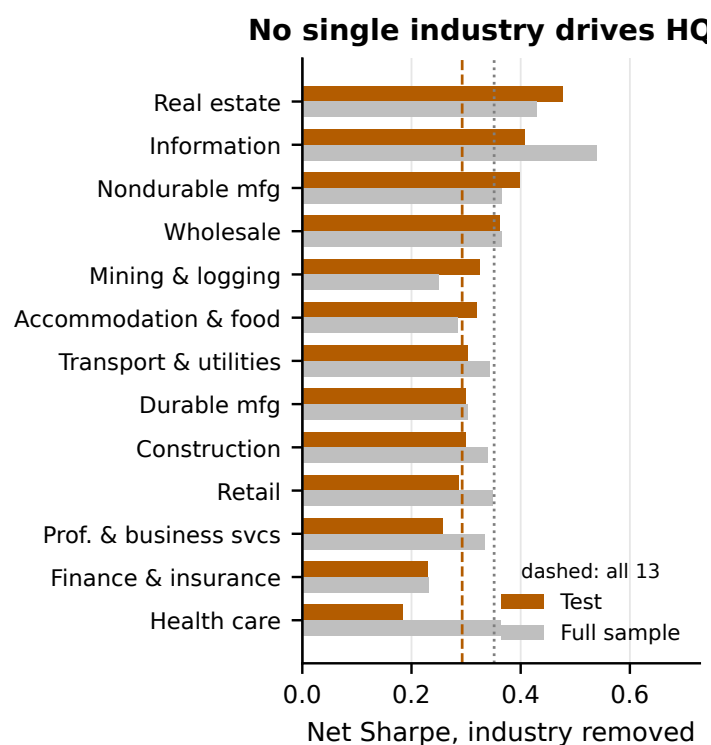


FIGURE 16. No single industry carries HQ12: removing any one leaves the test Sharpe between +0.18 and +0.48. Source: v3_r7_dropone.csv.

POST-HOC

statistic	Test Sharpe	Full Sharpe
minimum	+0.18	+0.23
median	+0.30	+0.34
maximum	+0.48	+0.54
all 13 groups (HQ12)	+0.29	+0.35
industry whose removal lowers test Sharpe most	Health care	
industry whose removal raises test Sharpe most	Real estate	

R8 Tightness moderator

post-hoc

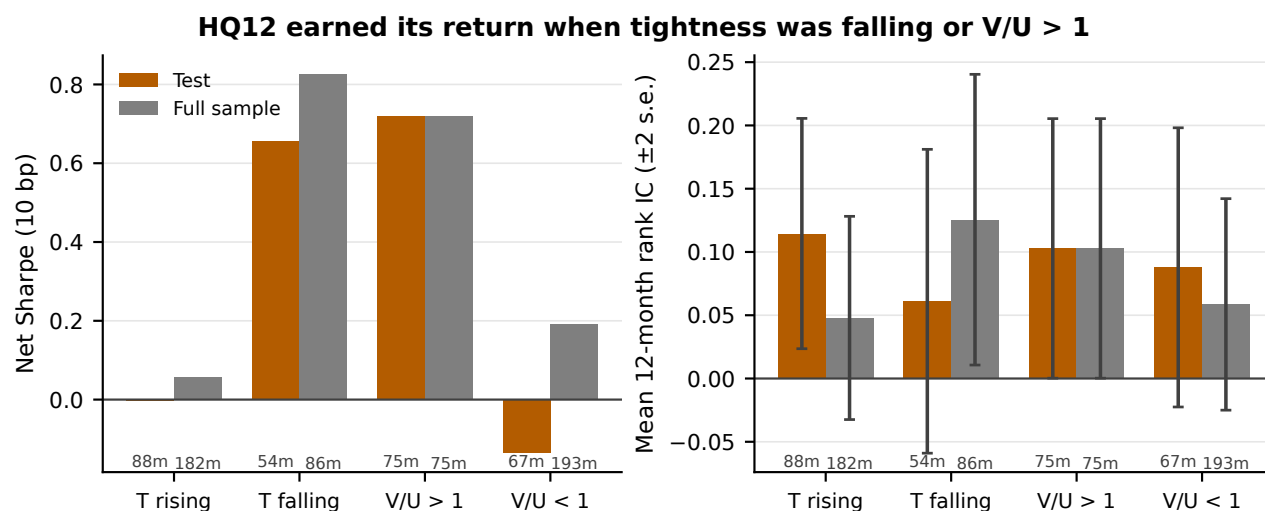


FIGURE 17. HQ12 earned its return when tightness was falling or $V/U > 1$; in tightening months the Sharpe was zero although the 12-month IC was highest. Source: v3_r8_tightness.csv.

POST-HOC

Months	Sharpe	Net p.a.	IC 12m (t)
88	-0.00	-0.0%	+0.115 (+2.52)
54	+0.66	+6.8%	+0.061 (+1.02)
75	+0.72	+6.0%	+0.103 (+2.00)
67	-0.14	-1.2%	+0.088 (+1.59)
182	+0.06	+0.4%	+0.048 (+1.19)
86	+0.83	+7.6%	+0.126 (+2.19)
75	+0.72	+6.0%	+0.103 (+2.00)
193	+0.19	+1.4%	+0.059 (+1.40)

R9 Block-bootstrap interval

post-hoc

Window	Sharpe	90% interval	Months	Excludes 0
test	+0.29	[-0.07, +0.65]	142	no
full	+0.35	[+0.05, +0.64]	268	yes

R10 As-known data

post-hoc

check	status	note
R10 as-known (ALFRED vin- tage) rerun of W-HQ12	not run: no key	analysis/.env has no FRED_API_KEY; historical v2/v3 re- sults use revised data with fixed lags (data_vintage.md)

R11 Fundamental law, test window

confirmatory (A3) / post-hoc

Book (test window)	IC	Horizon	BR/yr	TC	Implied IR	Ceiling (TC=1)	Realised Sharpe
A3 primary	-0.021	1m	156	0.80	-0.21	-0.27	-0.58
W wage growth	+0.049	1m	156	0.91	+0.56	+0.62	+0.16
HQ12 -(hires+quits) 12m	+0.096	12m	13	0.48	+0.17	+0.35	+0.29

Deflated Sharpe at 112 and 132 trials**post-hoc**

Window	DSR (112 trials)	DSR (132 trials)	Benchmark Sharpe (132)
test	0.02	0.02	0.76
full	0.03	0.03	0.56
test	0.00	0.00	0.76
full	0.00	0.00	0.56
test	0.09	0.08	0.76
full	0.16	0.15	0.56
test	0.06	0.05	0.76
full	0.18	0.16	0.56

Reading rule**post-hoc**

condition	holds
R1 hires-only and quits-only test Sharpe > 0	yes
R2 test 12m IC $t \geq 2$ and 1m IC $t < 2$	yes
R3 holding 9 and 18 months test Sharpe > 0	yes
R4 both test halves Sharpe > 0	no
R5 test break-even cost > 25 bp	yes
R7 minimum drop-one test Sharpe > 0	yes
R9 90% interval excludes 0 (test or full)	yes

6 Study 1 robustness and Study 2.3

Study 1 rows labelled confirmatory are read from the saved results and are unchanged; everything else is post hoc (spec `analysis/specs/study2_3_robustness.md`, 162 looks). Sources: `analysis/output/study1/robustness.md` and `analysis/output/study2/robustness_2_3.md`.

Study 1: A3 and A0

confirmatory / **post-hoc**

Check (test window, as-known data unless stated)	A3 agent features (primary)	A0 fixed rule
Test Sharpe, data as known	−0.58	−0.36
Test Sharpe, data first release	−0.52	−0.16
Test Sharpe, data revised	−0.56	−0.26
Test Sharpe, data fixed lag	−0.53	−0.27
Net return p.a.	−4.6%	−3.5%
IC, one month (t)	−0.021 (−1.08)	−0.020 (−0.71)
Factor alpha per month (t)	−0.38% (−1.86)	−0.25% (−0.92)
Placebo p (circular shift)	0.72	n/a (primary only)
Holm-adjusted p, IC and return	1.00, 1.00 (A3 vs A1)	1.00, 1.00 (A1 vs A0)
Deflated Sharpe, research (41 trials)	0.003	0.000
Sharpe at 0 / 10 / 25 bp	−0.45 / −0.58 / −0.79	n/a / −0.36 / n/a
Sharpe, first test half	−0.72	−0.55
Sharpe, second test half	−0.49	−0.20
Sharpe, excluding Mar-Dec 2020	−0.51	−0.18
Sharpe, last 18 months	+0.85	−1.56
Drop one group, Sharpe range	−0.70 (Wholesale) to −0.24 (Durable mfg)	n/a (primary only)
Gates G1 timing / G2 prediction / G3 alpha / G4 robustness / G5 implementable	pass / fail / fail / fail / fail	n/a (primary only)
Verdict	Do not implement	not primary
Block-bootstrap 90% interval, test Sharpe	[−0.99, −0.16]	[−0.91, +0.20]
A3 score IC at 1 / 6 / 12 / 24 months (t)	−0.021 (−1.08) / −0.004 (−0.09) / −0.013 (−0.22) / −0.000 (−0.00)	n/a
A3 Sharpe with cap 2 / cap 3 / no cap	−0.58 / −0.61 / −0.62	n/a
A3 Sharpe, rank-weighted book	−0.49	n/a

check	ok	value
SHA-256 of FREEZE.json equals evaluation	True	60ea05294ffc856e
tion_started.freeze_sha256		
Freeze recorded (UTC)	True	2026-10-02T06:00:09.537758+00:00
Test opened (UTC)	True	2026-10-02T06:00:09.578162+00:00
Test opened after the freeze	True	0.040 s later
Files in the freeze inventory	True	525
of which present in the repository	True	338
present files whose hash matches the inventory	True	337
present files that differ	True	outputs/human_migration_review.md
First test decision in the freeze record	True	2014-10-31

A3 agent features (Study 1 primary)	HQ12 (Study 2, post-hoc)
−0.021 (−1.08)	−0.020 (−0.87)
−0.029 (−1.08)	+0.007 (+0.18)
−0.004 (−0.09)	+0.017 (+0.40)
−0.015 (−0.27)	+0.069 (+1.98)
−0.013 (−0.22)	+0.096 (+2.43)
+0.006 (+0.13)	+0.085 (+2.00)
−0.000 (−0.00)	+0.063 (+1.86)

R1 As-known rerun

post-hoc

Sharpe, fixed lag	Sharpe, as known	IC (t), fixed lag	IC (t), as known	Alpha t	Same top 3	Same bottom 3
+0.16	+0.13	+0.049 (+2.04)	+0.041 (+1.73)	+0.18	76%	70%
−0.15	+0.24	+0.042 (+1.65)	+0.021 (+0.75)	+0.71	0%	0%
+0.35	+0.46	+0.032 (+1.21)	+0.036 (+1.40)	+1.93	84%	74%
+0.29	+0.20	+0.096 (+2.43)	+0.085 (+2.26)	+0.92	39%	51%

HQ12 check	Fixed lag (Study 2.2)	As known (Study 2.3)
Test Sharpe (10 bp)	0.2928175594619901	0.20353671517398647
Full-sample Sharpe (10 bp)	0.3513864909801854	0.31225166932716536
12-month IC, test	0.09617481754886337	0.08493414981964602
12-month IC t, test	2.42513564294896	2.2556531861001075
1-month IC t, test	−0.8683774566556943	−0.4217069291085937
Component hires only, test Sharpe	0.18928236667462156	0.18906453181867547
Component quits only, test Sharpe	0.16038100316084597	0.16792187564407968
Holding 9 months, test Sharpe	0.14037101563145116	0.15547151432490372
Holding 18 months, test Sharpe	0.30203574184950205	0.10844526496839682
First test half, Sharpe	−0.20857332220438748	−0.23120954703996147
Second test half, Sharpe	0.6382650516963818	0.49169151023338104
Test Sharpe at 0 bp	0.34296559563402257	0.2545719035225398
Test Sharpe at 25 bp	0.2174979357122725	0.12683023369254018
Test Sharpe at 50 bp	0.09172033492538699	−0.001402237091266201
Break-even cost, test (bp)	68.295908026001	49.80577710413222
Drop one group, minimum test Sharpe	0.18405490659750104	0.047841537479403035
Drop one group, maximum test Sharpe	0.4765507360777538	0.36096206448969387
Bootstrap 90% interval, test: low	−0.0675866768012903	−0.14470332505298303
Bootstrap 90% interval, test: high	0.6498532971228607	0.5673841209775896
Alpha per month (%), test, HAC 12	0.26094534893481813	0.1564993705066555
Alpha t, test, HAC 12	1.480671711289157	0.9227023711499632
Bootstrap 90% interval, full: low	0.05035649038376843	0.024425284981418837
Bootstrap 90% interval, full: high	0.6414508516824874	0.6027188662984058
Alpha per month (%), full, HAC 12	0.2646302974901718	0.20962061867799037
Alpha t, full, HAC 12	1.9203541055109492	1.8124813869598275
Deflated Sharpe, test (132 looks fixed lag; 162 as known)	0.04937867555521646	0.021460135063184792

R2 Risk overlay and R3 rank-weighted books

post-hoc

Book	Cap	Sharpe	Realised vol	Mean gross	Cap binds	Max drawdown
A3	2	−0.58	7.8%	1.98	87%	−41.1%
A3	3	−0.61	9.7%	2.57	35%	−50.7%
A3	none	−0.62	10.3%	2.79	0%	−54.9%
HQ12	2	+0.29	8.7%	1.97	83%	−14.4%
HQ12	3	+0.24	10.5%	2.51	39%	−20.3%
HQ12	none	+0.24	11.1%	2.78	0%	−22.6%

book	unit_vol_p5	unit_vol_p50	unit_vol_p95	gross_needed_p5	gross_needed_p50	gross_needed_p95
A3	0.045	0.075	0.110	1.85	2.60	4.19
HQ12	0.033	0.064	0.103	1.79	2.45	4.24

Book	Weights	Test Sharpe	TC	Turnover	Implied IR
A3	top 3 / bottom 3	−0.58	0.80	0.91	−0.21
A3	rank-weighted	−0.49	0.83	0.86	−0.22
HQ12	top 3 / bottom 3	+0.29	0.48	0.37	+0.17
HQ12	rank-weighted	+0.31	0.49	0.37	+0.17

R4 Agents by seed, R5 tokens

post-hoc

Study	Arm	Seed	Valid	Best IC	Mean IC	Eligible	Regime-gated
Study 1 (Sonnet)	independent	1	17/18	+0.101	+0.014	2	29%
Study 1 (Sonnet)	independent	2	17/18	+0.102	−0.006		24%
Study 1 (Sonnet)	independent	3	13/18	+0.046	−0.014		15%
Study 1 (Sonnet)	islands	1	16/18	+0.052	−0.003	2	25%
Study 1 (Sonnet)	islands	2	17/18	+0.058	+0.005		6%
Study 1 (Sonnet)	islands	3	18/18	+0.037	−0.008		22%
Study 1b (Opus)	independent	1	18/18	+0.088	−0.002	1	28%
Study 1b (Opus)	independent	2	18/18	+0.045	−0.006		11%
Study 1b (Opus)	independent	3	18/18	+0.045	+0.002		6%
Study 1b (Opus)	islands	1	17/18	+0.125	+0.026	6	6%
Study 1b (Opus)	islands	2	18/18	+0.191	+0.049		39%
Study 1b (Opus)	islands	3	17/18	+0.091	+0.009		47%

study	metric	islands_minus_independent	largest_seed_range_within_arm	gap_within_seed_range	perm_p_two_sided	relabelings
Study 1 (Sonnet)	best_ic		−0.034	0.056	True	0.30 20
Study 1 (Sonnet)	mean_ic		−0.001	0.028	True	1.00 20
Study 1b (Opus)	best_ic		+0.076	0.099	True	0.10 20
Study 1b (Opus)	mean_ic		+0.030	0.040	True	0.10 20

study	arm	calls_with_usage	tokens_total	tokens_per_call	output_tokens_per_call
Study 1 (Sonnet)	independent		54	120,932	2,239
Study 1 (Sonnet)	islands		54	134,185	2,485
Study 1b (Opus)	independent		54	180,555	3,344
Study 1b (Opus)	islands		54	214,003	3,963

study	n	spearman_output_tokens_vs_ic	spearman_output_tokens_vs_abs_ic
Study 1 (Sonnet)	98	+0.13	+0.21
Study 1b (Opus)	106	+0.41	+0.24

R6 Effective breadth, R7 capacity

post-hoc

matrix	participation_ratio
13 group excess returns	2.0
13 group relative returns (group minus equal-weight mean)	9.1
z(W) across groups	7.4
HQ12 score across groups	10.1

book	binding_etf	binding_adv_usd_m	monthly_trade_share_of_aum	aum_at_1pct_adv_usd_m	aum_at_5pct_adv_usd_m	spy_adv_usd_bn	smallest_etf_adv_usd_m	smallest_etf
A3	PEJ	3.2	0.063	0.5	2.5	47.0	3.2	PEJ
HQ12	PEJ	3.2	0.027	1.2	5.9	47.0	3.2	PEJ

study	judge	pairs	defined	agree	judge_yes	human_yes	false_yes	missed_yes
Study 1 (Sonnet)	original judge	8	2	1	1	1	1	0
Study 1 (Sonnet)	judge with verbatim quote	8	8	7	2	1	1	0
Study 1 (Sonnet)	with quote, fenced JSON accepted (supplementary)	8	8	7	2	1	1	0
Study 1b (Opus)	original judge	17	17	10	10	3	7	0
Study 1b (Opus)	judge with verbatim quote	17	7	3	5	3	4	0
Study 1b (Opus)	with quote, fenced JSON accepted (supplementary)	17	17	9	11	3	8	0

condition	seeds	proposals	valid	regime_proposals	regime_share	nonmissing_below_half_share	informative_below_half_share	best_ic
A_with_tercile_split	3	54	48	4	8%		0%	8% +0.070
B_without_tercile_split	3	54	50	3	6%		0%	6% +0.070

Time patterns, turnover and costs

confirmatory / post-hoc

Strategy	Study	Test	Full sample	Post-2010	Last 18 months
A0 fixed rule	Study 1	−0.36	−0.48	−0.41	−1.56
A1 ridge	Study 1	−0.51	−0.52	−0.66	+0.73
A1-T ridge x tightness	Study 1	−0.59	−0.58	−0.73	−0.37
A3 agent features, independent (primary)	Study 1	−0.58	−0.44	−0.62	+0.85
A4 agent features, communicating	Study 1	−0.79	−0.56	−0.74	+1.23
W wage growth (declared headline)	Study 2	+0.16	+0.15	+0.12	−1.71
SC signed composite	Study 2	−0.15	−0.11	−0.16	−1.34
W+MOM wage growth + momentum	Study 2	+0.35	+0.32	+0.24	−0.78
HQ12 hires + quits reversal, 12-month hold	Study 2	+0.29	+0.35	+0.30	+0.29

Book (test window)	Turnover p.a.	Cost drag p.a.	Gross p.a.	Net p.a.	Break-even (bp)
A3 agent features, independent (primary)	11.0x	1.11%	−3.5%	−4.6%	−32
W wage growth (declared headline)	4.7x	0.47%	+2.1%	+1.7%	+46
SC signed composite	8.7x	0.88%	−0.5%	−1.3%	−5
W+MOM wage growth + momentum	5.8x	0.59%	+3.6%	+3.1%	+62
HQ12 hires + quits reversal, 12-month hold	4.4x	0.44%	+3.0%	+2.6%	+68
A3 rank-weighted	10.3x	1.05%	−2.5%	−3.6%	−24
HQ12 rank-weighted	4.4x	0.45%	+3.0%	+2.6%	+69

7 The agent experiment

study	item	value
Both	Arms	islands (communicating), independent (control)
Both	Agents per arm	Temporal, Composition, Interactions
Both	Proposals per agent x repetitions	6 x 3
Both	Migration reports	after proposals 2 and 4, at most 120 words
Both	Selection (repetition 1 only)	$t \geq 1.5$, same sign in both halves, $ \text{corr} < 0.8$, at most 3 per arm
Sonnet	Valid proposals	98 / 108
Sonnet	Migration reports admitted	7 / 18
Sonnet	Defined LLM-judge uptake labels	8 / 36
Sonnet	Human-audited judge pairs	8
Opus	Valid proposals	106 / 108
Opus	Migration reports admitted	15 / 18
Opus	Defined LLM-judge uptake labels	84 / 88
Opus	Human-audited judge pairs	17
Opus	Human-judge agreement	10 / 17 (human yes 3, judge yes 10)

Selected features and their research→test fate:

Study	Arm	Feature (decoded)	Research IC (n)	Test IC (n)
Sonnet	independent	regime(yoy(Earnings), "low")	+0.065 (48)	n/a (3)
Sonnet	independent	regime(spread(tsz(Layoffs, 60), tsz(Hires, 60)), "low")	+0.070 (43)	n/a (3)
Sonnet	islands	spread(chg(Layoffs, 3), chg(Openings, 3))	+0.037 (123)	−0.023 (142)
Sonnet	islands	spread(chg(Layoffs, 6), chg(Openings, 6))	+0.052 (120)	−0.004 (142)
Opus	independent	mult(regime(tsz(Hires, "exp"), "high"), csrank(tsz(Openings, "exp")))	+0.088 (15)	−0.028 (133)
Opus	islands	tsz(lchg(lag(Earnings, 1), 12), 36)	+0.064 (78)	+0.022 (142)
Opus	islands	regime(spread(chg(Quits, 3), chg(Hires, 3)), "low")	+0.125 (48)	n/a (3)
Opus	islands	tsz(lchg(lag(Hours, 1), 12), 36)	+0.057 (78)	+0.011 (142)

Human vs judge (Opus): 10/17 agreement; human yes 3, judge yes 10; all 7 disagreements judge-yes/human-no. Models from the logs: Sonnet 5.5 at medium effort for the confirmatory study (199 logged calls), Opus 5.5 at xhigh for the follow-up (294 calls); claim A06.

8 AI-interaction inventory

ID	Purpose	Model	Prompt file	What was right	What was wrong	What changed	Team critique
P1	Source code audit before research	claude-sonnet-5-5 (medium); Opus study: claude-opus-5-5 (xhigh)	outputs/follow_the_workers/outputs/p1_request.json	25 numbered findings; found the shared CES series for groups 9/10 and several metadata inconsistencies.	Several alleged timing errors were not supported (release filtering pre-cedes the month bound); no numeric change resulted.	Mapping meta-data corrected (prescribed vs verified scope); construction SA exception flagged; no basket or feature change.	
P2	Blinded signal-search agents (two arms, 3 repetitions)	claude-sonnet-5-5 (medium); Opus study: claude-opus-5-5 (xhigh)	params.py SYSTEM_PROMPT / TURN_PROMPT; outputs/agents/log.jsonl	Rediscovered wage growth (V05) blind in both studies; produced valid grammar expressions (98/108 Sonnet, 106/108 Opus).	Highest research t-statistics came from regime gates that were active in 3 test months; every selected feature failed the test.	Mechanical selection only; no engineer edits. v2 restores wage growth as a public, ungated signal.	
P3	Migration reports and LLM judge of idea uptake	claude-sonnet-5-5 (medium); Opus study: claude-opus-5-5 (xhigh)	outputs/agents/migrations.json; judge prompt in agents.py trace_migrations	Opus: 15/18 reports admitted and 84/88 usable labels; a human audit of 17 pairs was possible.	Sonnet: 28/36 judge outputs were not bare JSON; judge over-attributes uptake (10 yes vs human 3; all 7 disagreements judge-yes).	Judge output treated as an association, not evidence; human labels reported alongside.	
P4	Robustness red team on research tables	claude-sonnet-5-5 (medium); Opus study: claude-opus-5-5 (xhigh)	outputs/follow_the_workers/outputs/p4_request.json	Ten ranked concerns; called "do not implement" from research tables before the seal opened.	Asserted uncomputed p-values and proposed unregistered thresholds; every suggestion was already covered by a registered diagnostic.	No new test adopted (all dispositions "covered by" or "not adopted"); registered diagnostics retained.	
P5	Post-mortem review of the frozen study	claude-opus-5-5 (Claude Code, 3 Oct 2026)	conversation request (Piero, 3 Oct 2026); docs/FINDINGS_AND_PROPOSAL.md	Found the wrong A0 signs, the dropped wage signal, the 12-month hiring sign, the 3-of-142-month agent features, the one-bin tightness split.	Guessed the engineering assistant was OpenAI Codex (unverified); used a momentum definition that included month m-1; quoted W ICs with a non-frozen start date.	Momentum redefined on m-12..m-2; W ICs re-computed with the frozen feature start; Codex guess withdrawn (to be confirmed by Alex).	
P6	This iteration: rebuild, diagnostics, frozen v2/v3 specs, forward test, report	claude-opus-5-5 (Parts 1-3); claude-fable-5-1 (overnight)	three prompts (re-build and diagnostics; v2/v3 freeze, forward test and report; final repository pass) and a one-page rules file, kept outside the repository	Reproduced the sealed A0 returns to 1e-16; froze specs before running; reported all variants; stopped at a failed gate instead of working around it.	The forward spec assumed August JOLTS would not be out by 29 Sep; it was released that day, so the pre-committed check stopped the run.	Amendment 001 (fixed-lag rule, W+MOM/HQ12 added, log guarantees v2 outputs never changed).	

Output files: P1: p1_response.md, p1_review.md; P2: candidates.csv, agents/candidate_records.json;

P3: agents/migration_judgments.json, human_review_submission.md; P4: p4_response.md, p4_review.md; P5: docs/FINDINGS_AND_PROPOSAL.md, analysis/output/study1/diagnostics.md; P6: analysis/run.ipynb, analysis/output/*.md, results/, report/main.tex.

9 Claims register

144 claims, 144 verified. Source: `results/claims.csv`.

ID	Label	Claim	Numbers	Ver.
C01	CONF	The preregistered primary A3 earned a test Sharpe of -0.58 over 142 months (Nov 2014-Aug 2026).	-0.58; 142	y
C02	CONF	A3 lost 4.6% a year net of costs, with 7.8% volatility and a 41% maximum drawdown.	-4.6; 7.8; -41	y
C03	CONF	A3 one-month rank IC in the test was -0.021 (Newey-West $t = -1.08$).	-0.021; -1.08	y
C04	CONF	A3 factor alpha was -0.38% a month after FF5, momentum and industry momentum.	-0.38	y
C04b	CONF	The t-statistic of the A3 alpha was -1.86.	-1.86	y
C05	CONF	The circular-shift placebo p-value for A3 was 0.72.	0.72	y
C06	CONF	A3 break-even trading cost was -32 bp: it loses money even with free industry trading.	-32	y
C07	CONF	A3 passed gate G1 and failed G2, G3, G4 and G5.	True; False; False; False; False	y
C08	CONF	Test Sharpe ratios: A0 -0.36, A1 -0.51, A1-T -0.59, A3 -0.58, A4 -0.79.	-0.36; -0.51; -0.59; -0.58; -0.79	y
C09	CONF	Research-period Sharpe ratios (88 common months): A0 -0.44, A1 -0.54, A1-T -0.55, A3 -0.13, A4 -0.18.	88; -0.44; -0.54; -0.55; -0.13; -0.18	y
C10	CONF	A3 by window: first half -0.72, second half -0.49, excluding Mar-Dec 2020 -0.51, last 18 months +0.85.	-0.72; -0.49; -0.51; 0.85	y
C11	CONF	A3 descriptive course windows: full sample (from Jun 2007) -0.44, post-2010 -0.62.	-0.44; -0.62	y
C12	CONF	A3 Sharpe at 25 bp industry costs was -0.79 and at zero industry costs -0.45.	-0.79; -0.45	y
C13	CONF	A3 loads on profitability (RMW +0.19, $t = 2.2$) and against value (HML -0.15, $t = -2.0$).	0.19; 2.2; -0.15; -2.0	y
C14	CONF	A0 loads +0.22 on industry momentum ($t = 1.9$).	0.22; 1.9	y
C15	CONF	A3 ran at 1.1% a year of trading costs against a gross loss of 3.5% a year.	1.11; -3.5	y
C16	CONF	All four Holm-adjusted comparisons in the Sonnet study had adjusted $p = 1.0$.	1.0; 1.0; 1.0; 1.0	y
C17	CONF	Of the 142 test months, A3 was in the expanding-tercile "high" tightness bin in 133.	133	y
C18	CONF	The cross-validated ridge penalty was 1000, the top of the grid, for every learned model.	1000; 1000; 1000; 1000	y
C19	CONF	Of the agent features that passed research selection, 4 of 4 (Sonnet) and 6 of 7 (Opus) failed the frozen sealed criterion ($t \geq 1.5$ and positive IC in both test halves); the one Opus feature that passed it was not selected.	4; 4; 6; 7	y
C23	CONF	A1-T test: net return -4.9% a year, volatility 8.2%, maximum drawdown -36.3%.	-4.9; 8.2; -36.3	y
A06	CONF	Logged model calls: 199 in the confirmatory (Sonnet) study and 294 in the Opus follow-up (candidate, migration and judge calls; the evaluation entries in the log are not calls).	199; 294	y
A07	CONF	P1 audit: 25 numbered findings, 5 not supported by the files, 1 metadata fix, none a look-ahead error; the shared group-9/10 CES series had 584 nonmissing rows; the prompt body ran to 38,717 characters.	25; 5; 1; 584; 38717	y
A08	CONF	The blinding audit of all 199 Sonnet calls passed (108 candidate calls, 18 migration calls, 1 syntax repair).	199; 108; 18; 1; True	y
A09	CONF	Sonnet migration reports: 11 of 18 rejected, 10 of them for exceeding the 120-word cap (121 to 133 words).	18; 11; 10	y
A10	CONF	P4 red team: 6 of 10 concerns were covered by registered diagnostics; for the other 4 no additional test was adopted.	6; 4	y
V13	POST-HOC	On the research years alone the HQ12 12-month IC is +0.047 ($t = 0.90$) and its horizon-matched placebo p is 0.26.	0.047; 0.9; 0.26	y
V14	POST-HOC	The best post-hoc test Sharpe (W+MOM, +0.35) corresponds to t of about 1.2 over 142 months (Sharpe times $\sqrt{142/12}$).	1.2	y
F09	FORWARD TEST	The Q4 2026 book was first built at 2026-10-03T09:17:55Z (independent evidence: commit 3e8ecd7, 09:18:29Z, holds a byte-identical book_2026Q4.csv, SHA-256 857275...); book_first_build.sha256 records that first build and later notebook runs must reproduce it.	2026-10-03T09:17:55+00:00	y
X01	CONF	Opus follow-up: primary A4 test Sharpe +0.01, research Sharpe -0.40, placebo p 0.56, Sharpe at 25 bp -0.19.	0.01; -0.4; 0.56; -0.19	y
X02	CONF	Opus A4 one-month IC -0.002 ($t = -0.10$); alpha t 0.05; last 18 months Sharpe -1.41.	-0.002; -0.1; 0.05; -1.41	y

ID	Label	Claim	Numbers	Ver.
X03	CONF	Opus A3 test Sharpe -0.46.	-0.46	y
X04	POST-HOC	In the Opus A4 book, durable manufacturing alone contributed +0.27 of cumulative gross return in the test.	0.273	y
D01	POST-HOC	Wage growth W: rank IC +0.029 in research (t 1.54), +0.049 in test (t 2.04), +0.039 full sample (t 2.50).	0.029; 1.54; 0.049; 2.04; 0.039; 2.5	y
D02	POST-HOC	Full-sample one-month ICs: openings -0.005, hires +0.013, quits +0.009, layoffs +0.028, hours -0.010, A0 composite -0.013, industry momentum +0.020.	-0.005; 0.013; 0.009; 0.028; -0.01; -0.013; 0.02	y
D03	POST-HOC	Layoffs (F4) carry a positive IC (+0.033 research, +0.026 test) although A0 gave them a negative sign; hours (F5) carry a negative IC (-0.012, -0.007) although A0 gave them a positive sign.	0.033; 0.026; -0.012; -0.007	y
D04	POST-HOC	At a 12-month horizon hires have IC -0.062 (t -2.1) and quits -0.065 (t -1.7); at one month +0.013 and +0.009.	-0.062; -2.1; -0.065; -1.7; 0.013; 0.009	y
D05	POST-HOC	Wage growth IC by horizon: +0.039 at 1 month (t 2.5), +0.026 at 12 months.	0.039; 2.5; 0.026	y
D06	POST-HOC	Expanding tightness terciles put 133 of 142 test months in the "high" bin (3 low, 6 middle); V/U exceeded 1 in 0 research months and 75 test months.	133; 3; 6; 0; 75	y
D07	POST-HOC	A 12-month-change split is balanced: T rising in 93 research and 88 test months, falling in 32 and 54.	93; 88; 32; 54	y
D08	POST-HOC	Industry momentum 12-1 through the same backtest had a test Sharpe of +0.03.	0.03	y
D09	POST-HOC	A3 test P&L by industry: health care -26.2 pp, construction -19.2, finance -16.3, professional services -12.2; wholesale +14.9; trading costs -13.1.	-26.2; -19.2; -16.3; -12.2; 14.9; -13.1	y
D10	POST-HOC	Selected agent features: 2 of 4 regime-gated in each study; test IC undefined for 4 Sonnet and 2 Opus candidates; 0 (Sonnet) and 2 (Opus) selected features had a positive test IC.	2; 2; 4; 2; 0; 2	y
D11	POST-HOC	Correlation between research and test IC across valid agent candidates: 0.22 (Sonnet), 0.37 (Opus).	0.22; 0.37	y
D12	POST-HOC	Sonnet's two independent-arm features were active in only 3 of 142 test months; the best Opus island feature (research t 3.56) also in 3.	3; 3; 3.56; 3	y
D13	POST-HOC	The gross-exposure cap bound in 87% of A3 test months; ex-ante volatility averaged 7.5% against a 10% target (realised 7.8%).	87; 7.5; 7.8	y
D14	POST-HOC	Across the preregistered books the cap bound in 80% to 93% of test months.	80; 93	y
A01	CONF	Sonnet study: 98 of 108 proposals valid, 7 of 18 migration reports admitted, 8 of 36 judge labels usable, 8 pairs human-audited.	98 / 108; 7 / 18; 8 / 36; 8	y
A02	CONF	Opus study: 106 of 108 proposals valid, 15 of 18 reports admitted, 84 of 88 judge labels usable, 17 pairs human-audited, human-judge agreement 10 of 17 (human yes 3, judge yes 10).	106 / 108; 15 / 18; 84 / 88; 17; 10 / 17 (human yes 3, judge yes 10)	y
A03	CONF	All 7 human-judge disagreements were judge-yes / human-no (items 2, 3, 8, 12, 13, 16, 17).	7; 0	y
A04	CONF	Judge-labelled uptake of migrated ideas: 25% of usable Sonnet labels (2 of 8), 62% of Opus labels (52 of 84).	0.25; 0.619	y
A05	CONF	Best-so-far research IC, islands vs independent (mean over 3 runs): Sonnet 0.049 vs 0.083, Opus 0.136 vs 0.059; neither gap exceeds the seed-to-seed range.	0.049; 0.083; 0.136; 0.059; False; False	y
V_W__SR	POST-HOC	W net Sharpe (10 bp): research +0.09, test +0.16, full +0.15, post-2010 +0.12, last 18 months -1.71.	0.09; 0.16; 0.15; 0.12; -1.71	y
V_SC__SR	POST-HOC	SC net Sharpe (10 bp): research -0.06, test -0.15, full -0.11, post-2010 -0.16, last 18 months -1.34.	-0.06; -0.15; -0.11; -0.16; -1.34	y
V_W+MOM__SR	POST-HOC	W+MOM net Sharpe (10 bp): research +0.27, test +0.35, full +0.32, post-2010 +0.24, last 18 months -0.78.	0.27; 0.35; 0.32; 0.24; -0.78	y
V_HQ12__SR	POST-HOC	HQ12 net Sharpe (10 bp): research +0.50, test +0.29, full +0.35, post-2010 +0.30, last 18 months +0.29.	0.5; 0.29; 0.35; 0.3; 0.29	y
V01	POST-HOC	W test: one-month IC +0.049 (t 2.04), alpha t 0.29, placebo p 0.02, deflated Sharpe 0.02, Sharpe at 25 bp +0.10, turnover 0.39 per month.	0.049; 2.04; 0.29; 0.02; 0.02; 0.1; 0.39	y
V02	POST-HOC	HQ12 test: 12-month IC +0.096 (t 2.43), one-month IC -0.020 (t -0.87), alpha t 1.37, pre-declared placebo p 0.85, deflated Sharpe 0.06, Sharpe at 25 bp +0.22.	0.096; 2.43; -0.02; -0.87; 1.37; 0.85; 0.06; 0.22	y
V03	POST-HOC	HQ12 full sample: alpha t 1.96, 12-month IC +0.071 (t 2.06), deflated Sharpe 0.18, pre-declared placebo p 0.86.	1.96; 0.071; 2.06; 0.18; 0.86	y
V04	POST-HOC	HQ12 test: net return +2.6% a year, volatility 8.7%, maximum drawdown -14.4%, hit rate 55%, turnover 0.37 per month.	2.6; 8.7; -14.4; 55; 37	y
V05	POST-HOC	W+MOM test: alpha t 1.36, placebo p 0.26, deflated Sharpe 0.09.	1.36; 0.26; 0.09	y

ID	Label	Claim	Numbers	Ver.
V06	POST-HOC	SC test: one-month IC +0.042 (t 1.65) but alpha t -0.84.	0.042; 1.65; -0.84	y
V07	POST-HOC	No v2 variant met the pre-declared reading rule.	not distinguishable from noise after the looks already taken; not distinguishable from noise after the looks already taken; not distinguishable from noise after the looks already taken; not distinguishable from noise after the looks already taken	y
V08	POST-HOC	The highest deflated Sharpe among all v2 variants and windows was 0.18 (HQ12, full sample).	0.18	y
V09	POST-HOC	Horizon-matched (12-month IC) placebo for HQ12, added post-hoc: p 0.06 test, 0.04 full sample, 0.00 post-2010, 0.26 research.	0.06; 0.04; 0.0; 0.26	y
V10	POST-HOC	SC signs learned on research data: + for openings, hires, quits, layoffs and wages; - for hours.	1; 1; 1; 1; 1; -1	y
V11	POST-HOC	W test loadings: HML -0.43 (t -4.4), momentum +0.22 (t 3.3), SMB +0.28 (t 2.3).	-0.43; -4.4; 0.22; 3.3; 0.28; 2.3	y
F01	FORWARD TEST	WS Q4 2026 book: long construction, durable manufacturing and information; short mining, health care and accommodation & food.	0.3333333333333333; 0.3333333333333333; 0.3333333333333333; 0.3333333333333333; -0.3333333333333333; -0.3333333333333333	y - - -
F02	FORWARD TEST	HQ12 Q4 2026 book: long finance, real estate and accommodation & food; short mining, durable manufacturing and retail.	0.3333333333333333; 0.3333333333333333; 0.3333333333333333; 0.3333333333333333; -0.3333333333333333; -0.3333333333333333	y - - -
F03	FORWARD TEST	SPY hedge per unit tranche: W -0.40, SC -0.06, W+MOM -0.24, HQ12 -0.02.	-0.4; -0.06; -0.24; -0.02	y
F04	FORWARD TEST	W+MOM's long durable manufacturing and short professional services both map to XLI and cancel, so its tradable book has four legs.	0.0	y
F05	FORWARD TEST	The forward spec was frozen at 2026-10-03T09:03:56Z and amendment 001 at 2026-10-03T09:16:11Z, before the book was built (gate3.md records the order).	True; True	y
S01	POST-HOC	Our rebuilt A0 signal matches the frozen fixed-lag scores with pooled correlation 0.998.	0.998	y
S02	POST-HOC	As-known (ASOF) scores pick the same top three groups as our revised-data signal in only 42% of test months.	42	y
S03	POST-HOC	Durable manufacturing is 60% semiconductors (Chips) by market cap in August 2026; nondurable manufacturing is 60% pharmaceuticals.	Chips 60%, Mach 9%, Autos 7%; Drugs 60%, Hshld 9%, Soda 7%	y
S04	CONF	Gate 1: the 13 group returns match all 50 saved ledgers to 1.5e-16 and our estimators match results.json to 3.6e-15.	1.51e-16; 3.55e-15	y
C20	CONF	A3 under the four data variants: Sharpe -0.58 (as-known), -0.52 (first release), -0.56 (revised), -0.53 (fixed lag); IC -0.021, -0.040, -0.026, -0.021.	-0.58; -0.52; -0.56; -0.53; -0.021; -0.04; -0.026; -0.021	y
C21	CONF	A3 turned over 0.91 of its industry book a month in the test.	0.91	y
C22	CONF	The frozen power statement: 142 sealed months require roughly 0.6 annualised Sharpe for IID t of about 2.	142 sealed months require roughly 0.6 annualized Sharpe for IID t≈2; a null does not rule out a smaller effect.	y
S05	POST-HOC	Windows: research 125 months, test 142, full sample 268, post-2010 200, last 18 months 18.	125; 142; 268; 200; 18	y
F06	FORWARD TEST	SC Q4 2026 book: long durable manufacturing, retail and transport & utilities; short real estate, health care and accommodation & food.	0.3333333333333333; 0.3333333333333333; 0.3333333333333333; 0.3333333333333333; -0.3333333333333333; -0.3333333333333333	y - - -
F07	FORWARD TEST	W+MOM Q4 2026 book: long construction, durable manufacturing and finance; short retail, transport & utilities and professional services.	0.3333333333333333; 0.3333333333333333; 0.3333333333333333; 0.3333333333333333; -0.3333333333333333; -0.3333333333333333	y - - -

ID	Label	Claim	Numbers	Ver.
F08	FORWARD THESIS	Hashes: v2 spec ca658234..., forward spec 735a8ffa..., amendment 001 c76cde86..., v3 spec 312aef1d... (first 8 hex digits).	ca6582347904489b db6fac1496f895fd 18c91eba47ca2bc8 566e35cdf44a5832; 735a8ffa9c9c23fd 710dfae778a535ca d649685aa157dcde d3ae0e9996dfb21f; c76cde86a85ff703 d317cfeb15dc02a4 2e342d81a299da2a ab959adeb372b12b; 312aef1dd9551aa7 044d37ccf8562a8e 66cddfe741e0e109 7c15cff18ed8bfde	y
V12	POST-HOC	W turnover 0.39 and HQ12 turnover 0.37 of the industry book per month in the test; A3 0.91.	0.39; 0.37	y
X05	POST-HOC	Opus study: 7 island-arm candidates passed the research filter against 4 in the Sonnet study.	7; 4	y
R1a	POST-HOC	Held 12 months, hires alone earn a test Sharpe of +0.19 and quits alone +0.16; openings +0.42 and layoffs +0.16 (both with the same negative sign, for contrast).	0.19; 0.16; 0.42; 0.16	y
R1b	POST-HOC	Hires alone have a test 12-month IC of +0.093 (t 2.90); quits alone +0.085 (t 1.75).	0.093; 2.9; 0.085; 1.75	y
R2a	POST-HOC	Test-window IC of the HQ12 score by horizon: -0.020 at 1 month (t -0.87), +0.069 at 9 (t 1.98), +0.096 at 12 (t 2.43), +0.085 at 18 (t 2.00), +0.063 at 24 (t 1.86).	-0.02; -0.87; 0.069; 1.98; 0.096; 2.43; 0.085; 2.0; 0.063; 1.86	y
R2b	POST-HOC	The 12-month IC of the HQ12 score is +0.047 (t 0.90) in research and +0.071 (t 2.06) over the full sample.	0.047; 0.9; 0.071; 2.06	y
R3a	POST-HOC	Test Sharpe by holding period: +0.05 at 6 months, +0.14 at 9, +0.29 at 12, +0.30 at 18.	0.05; 0.14; 0.29; 0.3	y
R3b	POST-HOC	Research Sharpe by holding period: +0.31 at 6 months, +0.43 at 9, +0.50 at 12, +0.58 at 18.	0.31; 0.43; 0.5; 0.58	y
R4a	POST-HOC	HQ12 test Sharpe was -0.21 in the first half (71 months) and +0.64 in the second half.	-0.21; 71; 0.64	y
R4b	POST-HOC	HQ12 test Sharpe excluding Mar 2020-Dec 2021: +0.37; pre-2020: +0.14; 2020 onward: +0.37. Full-sample halves: +0.33 and +0.38.	0.37; 0.14; 0.37; 0.33; 0.38	y
R4c	POST-HOC	The 12-month IC of HQ12 was positive in both test halves (+0.097 and +0.095) even though the first-half Sharpe was negative.	0.097; 0.095	y
R5a	POST-HOC	HQ12 test Sharpe: +0.34 at 0 bp, +0.29 at 10, +0.22 at 25, +0.09 at 50 bp; break-even industry cost 68 bp (69 bp full sample).	0.34; 0.29; 0.22; 0.09; 68; 69	y
R6a	POST-HOC	With 12 HAC lags, HQ12 alpha is +0.26% a month in the test and +0.26% over the full sample.	0.26; 0.26	y
R6a2	POST-HOC	The t-statistics of the HQ12 alpha with 12 HAC lags are 1.48 (test) and 1.92 (full sample).	1.48; 1.92	y
R6b	POST-HOC	HQ12 loads against profitability in the test: RMW -0.36 (t -4.45) and SMB -0.17 (t -2.24); market, momentum and industry-momentum loadings are small.	-0.36; -4.45; -0.17; -2.24	y
R7a	POST-HOC	Dropping any one industry leaves the HQ12 test Sharpe between +0.18 (without health care) and +0.48 (without real estate), median +0.30.	0.18; 0.3; 0.48; Health care; Real estate	y
R8a	POST-HOC	HQ12 net Sharpe by regime at the decision (test): -0.00 when T rose over 12 months, +0.66 when it fell; +0.72 when V/U > 1, -0.14 when V/U < 1.	-0.0; 0.66; 0.72; -0.14	y
R8b	POST-HOC	The 12-month IC of HQ12 is highest when T is rising (+0.115, t 2.52) although the Sharpe in those months is zero.	0.115; 2.52	y
R9a	POST-HOC	Block-bootstrap 90% interval for the HQ12 Sharpe: [-0.07, +0.65] in the test (includes zero) and [+0.05, +0.64] over the full sample (excludes zero).	-0.07; 0.65; 0.05; 0.64	y
R11a	POST-HOC	Fundamental law, test window: A3 IC -0.021 x sqrt(156) x TC 0.80 implies IR -0.21 (realised -0.58); W IC +0.049 x sqrt(156) x TC 0.91 implies +0.56 (realised +0.16); HQ12 IC +0.096 x sqrt(13) x TC 0.48 implies +0.17, ceiling +0.35 (realised +0.29).	-0.021; 0.8; -0.21; -0.58; 0.049; 0.91; 0.56; 0.16; 0.096; 0.48; 0.17; 0.35; 0.29	y
R12	POST-HOC	With N_trials raised to 132, the HQ12 deflated Sharpe is 0.05 in the test and 0.16 over the full sample; the benchmark Sharpe that 132 null trials would produce is 0.76 (test) and 0.56 (full).	0.05; 0.16; 0.76; 0.56	y
R13	POST-HOC	Against the pre-declared battery rule, HQ12 fails only R4 (both test halves positive); R1, R2, R3, R5, R7 and R9 hold.	no; yes; yes; yes; yes; yes; yes	y

ID	Label	Claim	Numbers	Ver.
R10	POST-HOC	R10 (as-known ALFRED rerun) was not run: no FRED API key.	not run: no key	y
B01	CONF	Study 1 freeze record: the SHA-256 of FREEZE.json equals the hash stored in evaluation_started.json, and the test was opened after the freeze (0.040 s later, 2 Oct 2026, 06:00:09 UTC).	True; True; 0.040 s later	y
B02	CONF	Of the 525 files in the freeze inventory, 338 are in the repository and 337 match their frozen hash; the one that differs is the human migration review, written after the test.	525; 338; 337; outputs/human_migration_review.md	y
B03	POST-HOC	Block-bootstrap 90% interval for the test Sharpe (12-month blocks, 5,000 draws): A3 [-0.99, -0.16], A0 [-0.91, +0.20].	[-0.99, -0.16]; [-0.91, +0.20]	y
B04	POST-HOC	The A3 score has no IC at any horizon in the test: -0.021 (t -1.08) at 1 month, -0.004 (t -0.09) at 6, -0.013 (t -0.22) at 12, -0.000 (t -0.00) at 24.	-0.021; -1.08; -0.004; -0.09; -0.013; -0.22; -0.0; -0.0	y
B05	POST-HOC	A3 test Sharpe with the gross cap at 2, at 3 and with no cap: -0.58, -0.61, -0.62; realised volatility 7.8%, 9.7%, 10.3%; the cap binds in 87% and 35% of months.	-0.58; -0.61; -0.62	y
B05b	POST-HOC	A3 realised volatility by cap: 7.8% (cap 2), 9.7% (cap 3), 10.3% (no cap); cap binding in 87% and 35% of test months.	7.8; 9.7; 10.3; 87; 35	y
B06	POST-HOC	The rank-weighted A3 book earns a test Sharpe of -0.49 with a transfer coefficient of 0.83 (top 3 / bottom 3: -0.58, 0.80).	-0.49; 0.83; -0.58; 0.8	y
B07	CONF	A0 (fixed rule) test Sharpe by data version: -0.36 as known, -0.16 first release, -0.26 revised, -0.27 fixed lag.	-0.36; -0.16; -0.26; -0.27	y
B08	CONF	A0 by window: first test half -0.55, second half -0.20, last 18 months -1.56; factor alpha -0.25% a month (t -0.92).	-0.55; -0.20; -1.56; -0.25% (-0.92)	y
B09	CONF	Dropping one group at a time, A3 test Sharpe ranges from -0.70 (without wholesale) to -0.24 (without durable manufacturing).	-0.70 (Wholesale) to -0.24 (Durable mfg)	y
B10	CONF	Study 1 research-period deflated Sharpe (41 nominal trials): A3 0.003, A0 0.000.	0.003; 0.000	y
T01	POST-HOC	R1 validation: raw features rebuilt from the as-known panel equal Study 1 saved features exactly (maximum difference 0), and the as-known A0 score equals the saved one to 4e-16.	0; 0; 0	y
T02	POST-HOC	As-known test Sharpe: W +0.13, SC +0.24, W+MOM +0.46, HQ12 +0.20 (fixed lag: +0.16, -0.15, +0.35, +0.29).	0.13; 0.24; 0.46; 0.2; 0.16; -0.15; 0.35; 0.29	y
T03	POST-HOC	HQ12 as known: 12-month test IC +0.085 (t 2.26) against +0.096 (t 2.43) with fixed lags; W one-month IC +0.041 (t 1.73) against +0.049 (t 2.04).	0.085; 2.26; 0.096; 2.43; 0.041; 1.73	y
T04	POST-HOC	Share of test months in which the as-known and fixed-lag scores pick the same top three and the same bottom three groups: HQ12 39% and 51%, W 76% and 70%, W+MOM 84% and 74%, SC 0% and 0%.	39; 51; 76; 70; 84; 74; 0; 0	y
T05	POST-HOC	HQ12 as known by window: research +0.54, test +0.20, full +0.31, post-2010 +0.23, last 18 months +0.14.	0.54; 0.2; 0.31; 0.23; 0.14	y
T06	POST-HOC	HQ12 battery as known: first test half -0.23, second +0.49; break-even cost 50 bp; drop-one test Sharpe between +0.05 and +0.36; hires only +0.19, quits only +0.17; holding 9 months +0.16, 18 months +0.11.	-0.23; 0.49; 50; 0.05; 0.36; 0.19; 0.17; 0.16; 0.11	y
T07	POST-HOC	HQ12 as known: bootstrap 90% interval [-0.14, +0.57] in the test and [+0.02, +0.60] over the full sample; alpha t 0.92 (test) and 1.81 (full sample, 12 HAC lags); deflated Sharpe 0.02 at 162 looks.	-0.14; 0.57; 0.02; 0.6; 0.92; 1.81; 0.02	y
T08	POST-HOC	As known, the last 18 months: W -1.31, SC +1.38, W+MOM -0.94, HQ12 +0.14; W+MOM test alpha t 1.93, the largest of the four.	-1.31; 1.38; -0.94; 0.14; 1.93	y
T09	POST-HOC	SC signs re-learned on as-known research data: negative for openings, hires, layoffs and hours, positive for quits and wages (with fixed lags only hours was negative).	-1; -1; 1; -1; -1; 1	y
T10	POST-HOC	Median gross exposure needed to reach 10% ex-ante volatility in the test: 2.60 for A3 and 2.45 for HQ12 (5th to 95th percentile 1.85 to 4.19 and 1.79 to 4.24); median unit-book volatility 7.5% and 6.4%.	2.6; 2.45; 1.85; 4.19; 1.79; 4.24	y
T10b	POST-HOC	Median ex-ante volatility of the unit book: 7.5% (A3) and 6.4% (HQ12).	7.5; 6.4	y
T11	POST-HOC	HQ12 test Sharpe with the cap at 2, 3 and none: +0.29, +0.24, +0.24; realised volatility 8.7%, 10.5%, 11.1%; the cap binds in 83% and 39% of months.	0.29; 0.24; 0.24	y
T11b	POST-HOC	HQ12 realised volatility by cap 8.7%, 10.5%, 11.1%; cap binding 83% and 39% of test months.	8.7; 10.5; 11.1; 83; 39	y
T12	POST-HOC	Rank-weighted HQ12: test Sharpe +0.31, transfer coefficient 0.49, monthly turnover 0.37 (top 3 / bottom 3: +0.29, 0.48, 0.37); fundamental-law implied IR +0.17 for both.	0.31; 0.49; 0.37; 0.29; 0.48; 0.17; 0.17	y

ID	Label	Claim	Numbers	Ver.
T13	POST-HOC	Islands minus independent, best research IC per seed: -0.034 in Study 1 (largest seed range 0.056, permutation p 0.30) and +0.076 in Study 1b (seed range 0.099, p 0.10); in both the gap is within the seed range.	-0.034; 0.056; 0.3; 0.076; y 0.099; 0.1; True; True	
T14	POST-HOC	Share of valid independent-arm proposals with a regime gate in Study 1: 29%, 24% and 15% across the three seeds.	29; 24; 15	y
T15	POST-HOC	Tokens per proposal call: 2,239 (independent) and 2,485 (islands) in Study 1; 3,344 and 3,963 in Study 1b. Spearman correlation between a proposal's output tokens and its research IC: +0.13 (Study 1), +0.41 (Study 1b).	2239; 2485; 3344; 3963; y 0.13; 0.41	
T16	POST-HOC	Participation ratio of the 13-group correlation matrix: 2.0 for excess returns, 9.1 for relative returns, 7.4 for z(W) and 10.1 for the HQ12 score.	2.0; 9.1; 7.4; 10.1	y
T17	POST-HOC	Capacity with the proxy ETFs: the binding ETF is PEJ (average daily volume 3.2million); monthly trading reaches 1% of it at 0.5 million of AUM for A3 and 1.2million for HQ12, and 5% at 2.5 and 5.9million; SPY trades 47 billion a day.	PEJ; 3.2; 0.5; 1.2; 2.5; 5.9; y 47	
T18	POST-HOC	Judge with a verbatim quote, Study 1 pairs (Sonnet): 8 of 8 answers usable (2 of 8 originally), agreement with the human 7 of 8, one false yes.	8; 2; 7; 1	y
T19	POST-HOC	Judge with a verbatim quote, Study 1b pairs (Opus): 7 of 17 answers were JSON only; accepting fenced JSON, agreement with the human is 9 of 17 (10 of 17 originally) with 8 false yes (7 originally) and no missed yes.	7; 9; 10; 8; 7; 0	y
T20	POST-HOC	Feedback ablation (independent arm, research data, 3 seeds, 54 proposals per condition): with the tightness-tercile split 4 of 48 valid proposals used a regime gate (8%); without it 3 of 50 (6%).	54; 48; 4; 8; 54; 50; 3; 6	y
T21	POST-HOC	The reconstructed research inputs of the ablation reproduce 94 of 98 saved Study 1 feedbacks exactly; the largest difference is 0.11.	98; 94; 0.11	y
T22	CONF / POST-HOC	Net Sharpe, full sample / post-2010 / last 18 months: A0 -0.48/-0.41/-1.56, A1 -0.52/-0.66/+0.73, A1-T -0.58/-0.73/-0.37, A3 -0.44/-0.62/+0.85, A4 -0.56/-0.74/+1.23 (Study 1); W +0.15/-0.12/-1.71, SC -0.11/-0.16/-1.34, W+MOM +0.32/+0.24/-0.78, HQ12 +0.35/+0.30/+0.29 (Study 2).	-0.48; -0.41; -1.56; -0.52; - y 0.66; 0.73; -0.58; -0.73; - 0.37; -0.44; -0.62; 0.85; - 0.56; -0.74; 1.23; 0.15; 0.12; -1.71; -0.11; -0.16; -1.34; 0.32; 0.24; -0.78; 0.35; 0.3; 0.29	
T23	CONF / POST-HOC	Annual turnover, cost drag and break-even cost in the test: A3 11.0x, 1.11%, -32 bp; W 4.7x, 0.47%, +46 bp; SC 8.7x, 0.88%, -5 bp; W+MOM 5.8x, 0.59%, +62 bp; HQ12 4.4x, 0.44%, +68 bp.	11.0; 1.11; -32; 4.7; 0.47; 46; y 8.7; 0.88; -5; 5.8; 0.59; 62; 4.4; 0.44; 68	
T24	POST-HOC	Rank-weighted books: A3 turnover 10.3x a year, break-even -24 bp; HQ12 4.4x, +69 bp.	10.3; -24; 4.4; 69	y
T25	POST-HOC	The Study 2.3 specification was hashed (068a3e03...) at 2026-10-04T09:18:39Z, before its first output; deflated Sharpe ratios in Study 2.3 use 162 looks.	068a3e0396332514 ee3b7667f31b9c1 ff6bfea7362d9e91 d69ecc013e791528	y

10 Known issues

- *W+MOM's ETF legs cancel.* Its long durable manufacturing and short professional services both map to XLI, so the tradable W+MOM book has four legs (book_2026Q4_etf.csv). Fix or disclosure: Disclose in the forward-test section; track W+MOM on French group returns as the primary record.
- *Revised vs as-known data.* Every historical v2/v3 number uses revised FRED values with fixed lags (option c in data_vintage.md). The frozen study found the same A3 IC under both rules, but as-known data picks the same top-3 groups in only 42% of months. Fix or disclosure: Label every v2/v3 number "revised data"; R10 (as-known rerun) is not run without a FRED key.
- *Overlap in the 12-month placebo.* The horizon-matched placebo for HQ12 uses overlapping 12-month returns; the circular shift preserves but does not correct the overlap, so its p-values are optimistic. Fix or disclosure: Keep the pre-declared 1-month placebo (p = 0.85) in the reading rule; report the 12-month one as a post-hoc check only.
- *Mapping dilution.* Labor data cover all establishments in an industry; returns cover listed

firms. Durable manufacturing is 60% semiconductors and nondurable manufacturing 60% pharmaceuticals by market cap. Fix or disclosure: State as a limit; name NAICS-based CRSP baskets as the follow-up.

- *Power.* 142 test months detect an annualised Sharpe of about 0.6 at $t = 2$. None of the post-hoc Sharpe ratios (0.1-0.35) is distinguishable from zero after deflation. Fix or disclosure: Say so in the limits; the forward test is a record, not a verdict.